

Theoretical limitations of multi-layer Transformer

Lijie Chen
UC Berkeley
Berkeley, United States
lijiechen@berkeley.edu

Binghui Peng
Google research
New York, United States
binghuip@google.com

Hongxun Wu
UC Berkeley
Berkeley, United States
wuhx@berkeley.edu

Abstract—Transformers, especially the decoder-only variants, are the backbone of most modern large language models. Yet, we have a very limited understanding of their limitations (i.e., what tasks they cannot solve) besides the simplest 1-layer case.

Due to the difficulty of analyzing multi-layer models, all previous work relies on unproven complexity conjectures to show limitations for multi-layer Transformers. In this work, we prove the first *unconditional* lower bound against multi-layer decoder-only transformers. For any constant L , we prove that any L -layer decoder-only transformer needs a polynomial model dimension ($n^{\Omega(1)}$) to perform sequential composition of L functions over an input of n tokens.

As a consequence, our results give: (1) the first depth-width trade-off for multi-layer transformers, exhibiting that the L -step composition task is exponentially harder for L -layer models compared to $(L + 1)$ -layer ones; (2) an unconditional separation between encoder and decoder, exhibiting a hard task for decoders that can be solved by an exponentially shallower and smaller encoder; (3) a provable advantage of chain-of-thought, exhibiting a task that becomes exponentially easier when the model is allowed to produce an intermediate sequence of tokens before outputting the final answer.

On the technical side, we propose the multi-party *autoregressive communication model* that abstracts the key aspects of a decoder-only Transformer. In particular, lower bounds within this communication model imply lower bounds against decoder-only transformers, independent of their implementation details. To prove lower bounds in this communication model, we also introduce a new proof technique that finds a certain *indistinguishable decomposition* of all possible inputs iteratively. We believe our new communication model and proof techniques will be helpful to understand the computational power of transformers further.

Index Terms—Large language model, Transformer, Computational complexity

I. INTRODUCTION

The Transformer architecture [58] forms the backbone of modern large language models (LLMs). When pre-trained on vast corpora and fine-tuned on expert datasets, Transformers achieve impressive performance across a

range of natural language tasks [1] and demonstrate emergent intelligence [10].

There is no doubt that the Transformer is an ingenious and powerful architecture, as evidenced by many of its substantial empirical successes. Still, a fundamental question remains elusive:

Does the architecture have any potential limitations or weaknesses?

Such limitations shed light on what types of tasks Transformers could struggle to solve in practice, as well as potential improvements on the underlying architectures that can be provably better than Transformers (if they overcome these weaknesses).

Ultimately, we hope to understand what tasks Transformers can or cannot *learn* to solve within the current training paradigm. This is challenging to study formally for two reasons: (1) a significant part of the success of LLM training is that they are trained on a gigantic dataset (i.e., the whole Internet), for which we do not have a good mathematical model. (2) Modern machine learning optimization algorithms (such as Adam [27] or AdamW [32]) are often applied to highly non-convex optimization tasks, making them difficult to analyze theoretically.

A representational power perspective on Transformers. A cleaner yet still fundamental question is to study what types of functions Transformers can or cannot *represent*: we say a Transformer can represent a function if one can set the parameters of the Transformer so that it computes this function. Clearly, if we show that Transformers cannot represent a particular function, they also cannot learn such a function, regardless of the training data and the optimization methods; that is, by studying its representational power, we bypass the difficulty in analyzing the optimization algorithms and the training data.

To study Transformers as a computational model, we often treat the context length (prompt length) n as a growing parameter and analyze how the size of the Transformer must scale to solve a given problem.

Lijie Chen is supported by a Miller Research Fellowship. Hongxun Wu is supported by Jelani Nelson's ONR grant N00014-24-1-2647.

Following the convention of [49], we say that a task is hard for Transformers if every Transformer that solves it must have size polynomial in n .

There has been extensive literature on the representation power of the simplest 1-layer Transformers: We know they cannot solve many basic tasks, such as the composition of two functions [40], 3SUM [49], and induction heads [50]. However, for the (arguably) most relevant case of multi-layer Transformers, unconditional lower bounds remain elusive even for two-layer models.¹ As a result, researchers have turned to study the computational power of Transformers by assuming certain conjectures from complexity theory. In particular, researchers have shown limitations of Transformers relying on assumptions on complexity classes such as TC⁰ [37], log-space computation L [40] and Massively Parallel Computation (MPC) [51]. These assumptions are however notoriously hard to prove, which brings the following fundamental question:

*Can we establish unconditional limitations
against multi-layer Transformers?*

Besides the natural motivation to remove unproven assumptions, resorting to complexity assumptions often loses the fine-grained structure of the Transformer model: these works often start by assuming a certain hard function is hard for a complexity class (such as TC⁰) that can simulate all constant-layer Transformers, which then implies this hard function cannot be computed by all constant-layer Transformers. It however fails to provide a more fine-grained understanding, such as the relative power between Transformers of different constant depths, as well as the relative power between encoder and decoder, since these lower bounds would apply to all of them equally. Jumping ahead, our proof techniques are fine-grained enough to provide such insights (distinguish between encoders and decoders, and between different depths).

A. Our result

In this paper, we address the question above in the affirmative by proving the first *unconditional* lower bounds against any constant layer (decoder-only) Transformer.²

¹There is even speculation in the literature that multi-layer Transformers face circuit lower bound barriers, meaning that proving lower bounds for multi-layer Transformers may require first resolving long-standing open questions on circuit lower bounds; see [21] for a detailed discussion.

²The original Transformer paper [58] consists of two main components: the *encoder* and the *decoder*. The encoder processes input tokens with pairwise attention mechanisms, capturing contextual relationships across the input. In contrast, the decoder employs causal masking to attend only to previous tokens, enabling autoregressive sequence generation. Following the success of GPTs [34], [45], all prominent large language models, including Claude, Gemini and Grok, have adopted the decoder-only approach.

Indeed, we prove that no small Transformer can solve sequential composition tasks over long context.

Theorem 1.1 (Lower bound for multi-layer Transformer). *Let H be the number of attention heads, d be the head dimension, p be the precision, L be the number of layers, n be the prompt length. For any $L \leq \tilde{O}(\log \log(n))$, an L -layer decoder-only Transformer could not solve L -sequential function composition whenever $Hdp \leq n^{2^{-4L}}$.*

A formal definition of the L -sequential function composition can be found at Definition II.1. Roughly speaking, given (the truth-table of) L functions $f_1, \dots, f_L$ and a query $w = (w_1, \dots, w_L)$, it asks to compute $i_1 = f_1(w_1), i_2 = f_2(w_2, i_1), \dots, i_L = f_L(w_L, i_{L-1})$. Intuitively, this task is hard for autoregressive decoders because when the decoder reads all the input functions $f_1, \dots, f_L$, it does not have any information about the query w due to its autoregressive nature; essentially, this means that all the compute the decoder spends reading $f_1, \dots, f_L$ is wasted and it must solve the task from scratch while reading the query w , which we prove to be impossible for small Transformers. We refer the reader to Section I-E for a discussion of why our lower bound only holds up for depth $\tilde{O}(\log \log(n))$.

Our L -sequential function composition can be interpreted as a generalization of the classical *pointer chasing* problem [8], [39], in which one is given (the truth-table of) m functions $f_1, f_2, \dots, f_m$ and an integer s , the goal is to compute $f_m(\dots(f_2(f_1(s))))$. The difference between our L -sequential function composition problem and the pointer chasing problem is that our functions f_i also take another input w_i in addition to the output of f_{i-1} . We also remark that [8] proved a strong multi-pass streaming lower bounds against solving such pointer chasing problem; this does not directly imply any multi-layer decoder lower bound since multi-pass streaming algorithms cannot efficiently simulate multi-layer decoders.

On the circuit lower bound barrier. It was suggested in [21] that unconditional lower bounds against *encoder-only* transformer would imply breakthrough circuit lower bounds.³ Our result avoids this barrier by exploiting the information bottleneck and autoregressive nature of *decoder-only* models. Namely, our proof crucially depends on the fact that in decoder-only models, each token cannot attend to any token after it.

1) *Composition and depth in Transformers:* “Compositional tasks” often refer to complex tasks that require a model to combine simpler skills or pieces of knowledge

³In [21] it was mentioned briefly in a footnote without providing a detailed argument; in Section B, we provide a detailed argument showing that an n^c -size lower bounds against encoders would imply breakthrough circuit lower bounds.

to solve. The ability to perform compositional tasks has been a central focus of empirical research [7], [15], [26], [42], [43], [59], [60], [64], [67], as compositionality is essential for reasoning and handling complex tasks. [15] demonstrated through extensive experiments that Transformers struggle with tasks requiring the sequential composition of elementary steps, such as multiplying multi-digit integers and solving logical puzzles, with performance rapidly declining as the depth of composition increases.

Our sequential function composition function can be seen as a simple example of sequential compositional tasks, where the model needs to combine knowledge from multiple functions $f_1, \dots, f_L$ sequentially to obtain the final output. Hence, our main result Theorem I.1 provides theoretical justification for the limitations of Transformers in executing sequential composition.

Indeed, Theorem I.1 highlights the importance of depth in Transformers. As we will elaborate in Section I-B, Theorem I.1 implies a depth-size tradeoff for Transformers in compositional tasks. This aligns with the empirical findings of [67], which demonstrate that depth plays a more critical role than width in reasoning and composition tasks.

B. Implications

Next, we discuss several implications of our main Theorem I.1.

1) *Depth-size tradeoff for Transformer*: As mentioned above, Theorem I.1 highlights the importance of depth in Transformers. Indeed, it implies an exponential depth-size tradeoff for Transformer architecture. This is because the L -sequential function composition can be easily solved by an $(L+1)$ -layer decoder-only Transformer, with only polylogarithmic number of parameters (i.e., $H = O(1)$, $d = \text{poly} \log(n)$, $p = \log(n)$).

Corollary I.2 (Depth-size tradeoff). *For any constant $L \geq 1$, there exists a task (a.k.a. L -sequential function composition) such that (1) an $(L+1)$ -layer Transformer can solve the task with polylogarithmic number of parameters while (2) any L -layer Transformer needs polynomial number of parameters to solve the task.*

The lower bound on induction heads in [50] gives a strong separation between the depth-1 and depth-2 transformers. Our results are the first to extend such separations between depth- L and depth- $L+1$ transformers for every constant $L \geq 2$.

a) *Prior work: depth-size tradeoff for neural networks*: Depth-size (or depth-width) tradeoff has been extensively studied for neural networks [9], [14], [17], [28], [30], [33], [35], [47], [48], [54], [56], [57], [66]. Most work has focused on the feed-forward ReLU network. However, as pointed out by [56], [57], proving

depth-width tradeoff for $L \geq 4$ layer ReLU network faces circuit lower bound and natural proof barriers for benign functions (i.e., functions that can be computed in polynomial time and has polynomial-bounded Lipschitz constant). Hence, existing work [14], [17] either focus on very shallow network ($L = 2, 3$), or construct separations based on high oscillatory function with super-polynomial Lipschitz constant [54].

2) *Separation between Transformer encoder/decoder*: Theorem I.1 also implies a separation between the encoder and the decoder architecture, since the L -sequential function composition task can be easily solved by an $O(\log(L))$ -layer Transformer encoder.

Corollary I.3 (Separation between encoder and decoder). *For any constant $L \geq 1$, there exists a task (a.k.a. L -sequential function composition) such that (1) an $O(\log(L))$ -layer Transformer encoder could solve the task with polylogarithmic number of parameters while (2) any L -layer Transformer decoder needs polynomial number of parameters to solve the task.*

There has been a lot of work [2], [3], [20], [38], [44] comparing the empirical performance of encoder and decoder architecture, see [53] for a detailed coverage. A recent work [18] compares the expressive power between encoder and decoder, showing a strong separation by assuming a conjecture of the hardness of a certain triplet counting problem. In contrast, our work gives the first unconditional separation without any unproven assumptions.

3) *Provable benefits of chain of thought*: The chain of thought (CoT) [61] is known to help with the reasoning by inducing the LLM to perform step by step reasoning and eventually leading to the correct answer. Formally, CoT allows the model to output an intermediate sequence of tokens (i.e., the reasoning steps) before outputs the final answer.

Theorem I.1 implies the first provable benefits of CoT, without relying on any computational complexity conjecture. This is because L -sequential function composition can be easily solved using L -steps of CoT with only polylogarithmic number of parameters.

Corollary I.4 (Provable benefits of CoT). *For any constant $L \geq 1$, there exists a task (a.k.a. L -sequential function composition) such that (1) an one-layer Transformer with CoT could solve the task with polylogarithmic number of parameters while (2) any L -layer Transformer decoder needs polynomial number of parameters to solve the task.*

a) *Prior work: benefits of CoT under complexity conjectures*: Previous work [19], [29], [36], [41] have proved that log-precision Transformer with CoT could simulate any polynomial-time algorithm with a

polynomially long CoT. Therefore, by further assuming certain complexity conjecture (e.g. $P \not\subseteq TC^0$), their results imply that constant depth Transformer with CoT could simulate poly-time algorithm, while constant depth Transformer ($\subseteq TC^0$) itself can not solve P-complete task.

Compare to the prior work, our lower bounds: (1) hold for a simpler function, i.e., the L -sequential function composition compared to the P-complete or NC^1 -complete tasks considered in prior works; (2) require a much shorter CoT length (L steps) compared to the polynomial steps required by prior works; and (3) hold unconditionally without any unproven complexity conjecture.

C. Technique overview

Below, we provide an overview of Theorem I.1. In Section I-C1, we briefly sketch the attention mechanism and the architecture of transformers. In Section I-C2, we introduce the autoregressive communication model, which captures the computation of decoder-only transformers. Then, in Section I-C3, we define the L -sequential function composition task, for which we will show hardness against decoder-only models. Finally, in Section I-C4, we sketch the proof that L -sequential function composition is hard in the autoregressive communication model, which implies Theorem I.1.

a) *Notation.*: We write $[n] = \{1, 2, \dots, n\}$ and $[n_1 : n_2] = \{n_1, n_1 + 1, \dots, n_2\}$. In the corner case, we let $[0] = \emptyset$.

1) *Attention mechanism and Transformers.*: We begin our overview with a brief introduction to the attention mechanism and Transformers. Readers familiar with Transformers may skip directly to Section I-C2. For a more formal treatment, please refer to Section II-A. For simplicity, we present only a single-head Transformer here.

First, the input to a Transformer, often called a prompt, is a string such as:

Q: How many r's are there in the word strawberry?
A:

To store this input, the string is divided into n tokens (each representing a constant number of characters). The i -th token is then embedded as a d -dimensional vector $x_i \in \mathbb{R}^d$. Here, we assume that we use only p bits of precision to store each real number, typically with $p = \Theta(\log(n))$. In modern Transformers, d is usually a large constant such as 1024.

We set $x^{(0)} = x$ as the input tokens. The Transformer processes the input by alternating between two types of layers, for any $\ell \in [L]$:

- **Attention layer** $f_{\text{attn}}^{(\ell)}$. The input and output to this layer are length- n sequences of d dimensional

vectors $x^{(\ell-1)}, y^{(\ell)} \in (\mathbb{R}^d)^n$. The layer encodes each input vector $x_i^{(\ell-1)}$ via three learned linear transformations $K^{(\ell)}, Q^{(\ell)}, V^{(\ell)}$ ⁴, into a key vector $k_i \in \mathbb{R}^d$, a query vector $q_i \in \mathbb{R}^d$, and a value vector $v_i \in \mathbb{R}^d$.

Then, for each position $b \in [n]$, the layer attends to all positions $a \leq b$, computing the output:

$$y_b^{(\ell)} = \frac{\sum_{a \leq b} \exp(\langle k_a, q_b \rangle) \cdot v_a}{\sum_{a \leq b} \exp(\langle k_a, q_b \rangle)}.$$

Here, the inner product $\langle k_a, q_b \rangle$ is called the *attention score*, measuring how relevant information at position a is to position b . The output $y_b^{(\ell)}$ is a weighted average of the value vectors v_a , where weights are determined by the softmax of the attention scores.

- **MLP layer** $f_{\text{mlp}}^{(\ell)}$. The MLP layer is an arbitrary function $g^{(\ell)} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ (usually implemented as a few layers of feedforward neural networks in practice) applied independently to each position:

$$x_i^{(\ell)} = g^{(\ell)}(y_i^{(\ell)}), \quad \text{for all positions } i \in [n].$$

The final network is just

$$f_{\text{tran}} = f_{\text{mlp}}^{(L)} \circ f_{\text{attn}}^{(L)} \circ \dots \circ f_{\text{mlp}}^{(1)} \circ f_{\text{attn}}^{(1)}.$$

These intermediate inputs and outputs, $x^{(\ell)}$ and $y^{(\ell)}$, are called *hidden states*. At the inference time, we only care about the last predicted token $x_n^{(L)}$ (hence in the above example, $x_n^{(L)}$ should equal the embedding of the answer token ‘3’). For a pictorial illustration, please refer to Figure 1.

In this simplified presentation, we omitted details such as residual connections, normalization layers, and positional encodings. However, these will be captured by our new autoregressive communication model and thus be captured by our lower bound.

2) *Autoregressive Communication Model.*: We now introduce our autoregressive communication model, which is a key conceptual contribution of this work. We begin by briefly reviewing existing communication models and highlighting their limitations.

a) *Transformer as a Communication Model.*: To analyze the limitations of Transformer architectures, previous works [40], [49], [51] proposed an insightful perspective by formulating it as a communication game. In this framework, any lower bound established for the communication game directly translates into a lower bound for Transformer architectures. Recall that for simplicity, we only present single-head case here in the overview.

Suppose the input x contains n tokens, each embedded as a vector $x_i \in \mathbb{R}^d$. We partition these tokens among N

⁴These transformations are shared across all positions i .

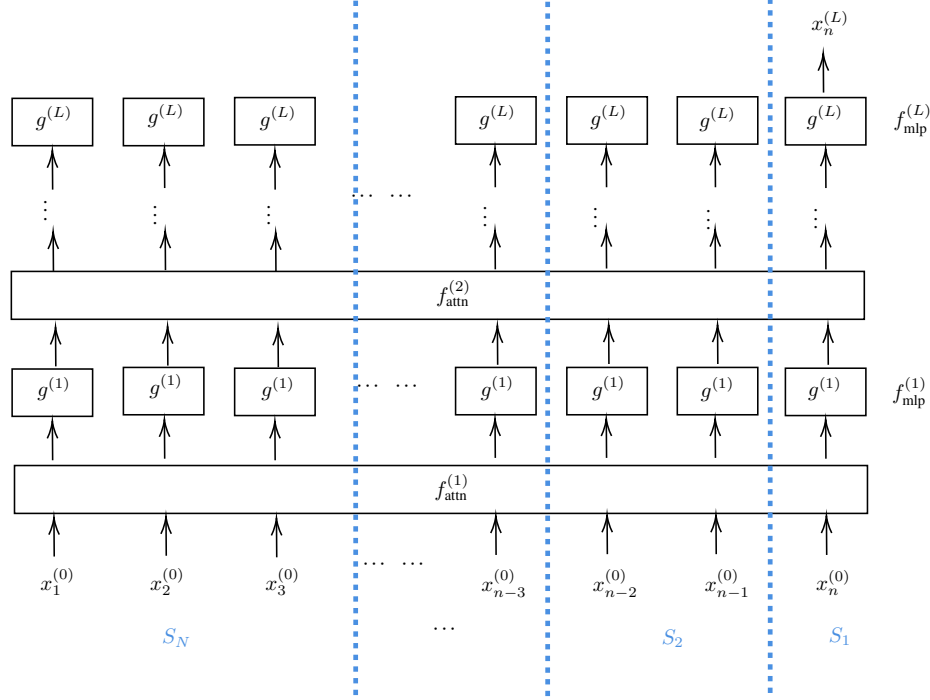


Fig. 1: The Transformer architecture. The blue dashed lines are for Section I-C2 describing how we divide it into N players $N, N-1, \dots, 1$ in our autoregressive communication model where each player i holds a set of positions S_i .

players, each player $i \in [N]$ privately holding a disjoint subset $S_i \subseteq [n]$, satisfying $S_1 \cup \dots \cup S_N = [n]$. Note that each subset S_i is not necessarily a contiguous interval.

The players' goal is to simulate an L -layer Transformer f_{tran} on input x through communication:

- Initially ($r = 0$), player i holds the input tokens at positions in S_i .
- After round r , each player i should hold the layer- r hidden state $x_a^{(r)}$'s corresponding to positions $a \in S_i$.
- Ultimately, after L rounds, the designated *output player* e , holding the last token ($n \in S_e$), outputs the token $x_n^{(L)}$.

At each round r , players simulate the attention mechanism and the MLP operations of layer r . The MLP operation can be applied independently at each position without communication. We focus here on simulating attention.

When player i computes the hidden state for a single position $b \in S_i$, it must consider all positions $a < b$ held by other players. Player i first sends a query message $\Gamma_{i,j} = q_b$ to every other player j . Player j responds

with a pair of partial sums

$$\Pi_{j,i} = \left(\sum_{a \in S_j, a < b} \exp(\langle k_a, q_b \rangle) \cdot v_a, \sum_{a \in S_j, a < b} \exp(\langle k_a, q_b \rangle) \right).$$

Player i aggregates these responses to calculate the new hidden state:

$$y_b^{(\ell)} = \frac{\sum_{j \in [N], j \neq i} \sum_{a \in S_j, a < b} \exp(\langle k_a, q_b \rangle) \cdot v_a}{\sum_{j \in [N], j \neq i} \sum_{a \in S_j, a < b} \exp(\langle k_a, q_b \rangle)}.$$

This exchange requires only $O(dp)$ bits per player pair and position b , independent of subset size.

b) Limitations for Multi-Layer Transformers.: At first glance, the approach above seems suitable for proving lower bounds for L -layer Transformers by establishing lower bounds over L communication rounds. However, prior research has only successfully derived lower bounds for single-layer ($L = 1$) Transformers due to the following limitation:

When $L = 1$, lower bounds are possible because we can partition tokens such that the output player e holds only the last token ($S_e = n$), requiring communication with all other players totaling $O(N \cdot dp)$ bits. Choosing the number of players so that $N \cdot dp \ll n$ ensures total communication is far less than the prompt length,

preventing the output player from trivially gathering all inputs.

This strategy fails even when $L = 2$. In round one, consider the player with the most tokens (without loss of generality, player 1). Calculating attention between S_i and S_1 demands at least $|S_i| \cdot dp$ bits of communication. Hence the player i can then simply send its inputs directly to player 1. Consequently, player 1 can trivially compute the result alone and communicate the 1-bit answer to the output player in the second round, eliminating nontrivial lower bounds.

c) *Our Setting.*: To overcome this issue, we propose an autoregressive communication model. The model is structured similarly with N players but labeled in reversed order $N, N-1, \dots, 1$ (See Figure 1). Now each player holds a consecutive interval of tokens S_i . Due to such reversed ordering, player N holds the first token, and player 1 (the output player) holds the last tokens.⁵

The protocol unfolds over L epochs (analogous to Transformer layers). Let z_i denote player i 's private input. The objective is for player 1 (the output player) to compute a function $f(z_N, \dots, z_1)$ after the L -th epoch.

d) *Beyond the Limitation: Forgetful Communication.*: The key insight in our model is that, to simulate an attention layer, when player i sends the query message $\Gamma_{i,j} = \{k_b\}_{b \in S_i}$ to player $j > i$, player j does not need to remember this message $\Gamma_{i,j}$ after answering the query $\Pi_{j,i}$! Thus, we can weaken the communication model by requiring that player j must forget all the information in $\Gamma_{i,j}$.

This ensures that information only flows in one direction: from player j to player i (with $j > i$). Such one-way information flow is a defining property of decoder attentions (also called autoregressive attentions), and this is why we call our model *autoregressive communication*. As a result of such one-way information flow, we can prevent the player i holding the largest S_i (for our hard problem in Section I-C3, this is player N who holds a prefix of the input) from gathering all the private inputs and bypass the previous limitation.

Formally, denote by $X_i^{(\ell)}$ the information held by player i after epoch ℓ (with $X_i^{(0)}$ initially being player i 's input). Each epoch $\ell \in [1 : L]$ proceeds as follows:

- The player i sends a message $\Gamma_{i,j}^{(\ell)}$ to all players $j \in [i+1 : N]$.
- The player $j \in [i+1 : N]$, based on its own information $X_j^{(\ell-1)}$ and player i 's message $\Gamma_{i,j}^{(\ell)}$, it sends a message $\Pi_{j,i}^{(\ell)}$ to player i .

⁵Note since the players are arranged in the reverse order, player i would attend to player $j > i$.

- Finally, the player i updates its collection of information as

$$X_i^{(\ell)} := X_i^{(\ell-1)} \cup \bigcup_{j>i} \Pi_{j,i}^{(\ell)}.$$

That is, the information state of player i is updated to include all messages received from players $j \in [i+1 : N]$.

Finally, the player 1 returns an output based on its information state $X_1^{(L)}$.

The most salient feature of the autoregressive communication model is that the players are forgetful. That is, the player j does not remember anything sent from player $i \in [1 : j-1]$; see Section III for a formal definition of the autoregressive communication model.

e) *Transformer as Autoregressive Communication.*: Finally, let us sketch how this autoregressive communication model is sufficient for simulating a Transformer. This is very similar to the simulation scheme before. For the MLP layer, each player just apply it separately on each position. For the attention layer, the message $\Gamma_{i,j}^{(\ell)}$ contains the queries from positions in S_i to positions in S_j and the returned message pair $\Pi_{j,i}^{(\ell)}$ contains the partial sum of key/value like before. Moreover the size of $\Gamma_{i,j}^{(\ell)}$ and $\Pi_{j,i}^{(\ell)}$ is bounded by $2Hdp \cdot |z_i|$, that is, they are proportional to the input length of player i . We call $B := 2Hdp$ the message bits of the communication model; see Lemma III.1 for a detailed proof of the simulation of decoder-only transformers by autoregressive communication protocols.

3) *Sequential function composition*: Now we elaborate on the L -sequential function composition task and explain why an autoregressive communication protocol with small message bits $B = 2Hdp$ fails to solve it.

a) *Intuition.*: We first provide intuitions on what makes a task hard for autoregressive communication models. Intuitively, the player i has stronger communication power than player $j > i$, since player i could communicate to all players $[i+1 : N]$ and it remembers all their returned message in the information state. Therefore, the failure of an autoregressive communication protocol happens in the regime of $|z_N| \gg |z_{N-1}| \gg \dots \gg |z_1|$.⁶ On the other hand, in this regime, player j possess much more information than player i regard the entire sequence $(z_N, \dots, z_1)$ since its input has larger size and its communication capacity is larger (i.e., $|z_j| \cdot 2Hdp \gg |z_i| \cdot 2Hdp$). In order to avoid shortcut in the communication, we must make sure that player 1 holds important “secrets” that are crucial for *all* players $j \in [2 : N]$.

⁶Note that on the other extreme, when $|z_1| \geq |z_j|$ ($\forall j \in [2 : N]$), there exists a trivial communication protocol since the player j could send its input z_j to player 1.

To this end, consider the L -sequential function composition task. Let $m, n_1, \dots, n_{L-1}$ be parameters and $N_\ell = m \cdot \prod_{i=1}^\ell n_i$ for $\ell \in [0 : L]$.

b) *L-sequential function composition.*: Our task, $L\text{-FuncComp}(w, z_0, z_1, \dots, z_L)$, takes a sequence of functions $z_0, z_1, \dots, z_L$ as input, where $z_0 \in [m]$ and $z_\ell : [N_{\ell-1}] \rightarrow [N_\ell]$ for $\ell \in [L]$ and a query $w = (w_1, \dots, w_{L-1}) \in [n_1] \times \dots \times [n_{L-1}]$. The output is defined inductively as follows: First, one computes

$$i_0 = z_0 \in [m], \quad i_1 = z_1(i_0) \in [N_0]$$

and one inductively computes, for each $\ell = 1, 2, \dots, L-1$:

$$i_2 = z_2(w_1, i_1) \in [N_1], \quad \dots, \quad i_{\ell+1} = z_{\ell+1}(w_\ell, i_\ell) \in [N_\ell].$$

The final output is then defined as

$$L\text{-FuncComp}(w, z_0, z_1, \dots, z_L) = i_L.$$

In the autoregressive communication model, for the L -sequential function composition task, we have $N = L + 2$ parties. For the sake of exposition, we rename them as player $L, L-1, \dots, 0, -1$, where the player ℓ receives z_ℓ ($\ell \in [0 : L]$) and the player -1 receives the query $z_{-1} := w$. For simplicity, we can assume that the message $\Gamma_{i,j}^{(\ell)}$ is just the whole information state $X_i^{(\ell-1)}$ of player i at the end of epoch $\ell-1$.

4) *Communication lower bound.*: Next, we elaborate on the communication lower bound for the sequential function composition task, which is the main technical part of this paper. In this overview, we hide the precise choice of parameters and highlight the key ideas behind the proof.

For convenience, we set $A_\ell = [N_{\ell-1}]^{N_{\ell-1}}$ be the input domain of player ℓ ($\ell \in [1 : L]$). We also set $A_0 = [m]$ and $A_{-1} = [n_1] \times \dots \times [n_{L-1}]$ be the input domain of player 0 and player -1 , respectively.

a) *Indistinguishable decomposition.*: Our key idea is the concept of *indistinguishable decomposition*, which is two sets $R_{\geq \ell}$ and $Z_{< \ell}$, where $R_{\geq \ell}$ is a set of input assignments to players $[\ell : L]$ and $Z_{< \ell}$ is a set of input assignments to players $[-1 : \ell-1]$, such that for every possible inputs $z_{< \ell} \in Z_{< \ell}$, all assignments in $R_{\geq \ell}$ are indistinguishable to players $[-1 : \ell-1]$ on inputs $z_{< \ell}$ after ℓ epochs (because they lead to the same transcripts).

For technical reasons, we will further assume (1) $Z_{< \ell}$ is a product set $Z_{< \ell} := Z_{-1} \times Z_0 \times \dots \times Z_{\ell-1}$, where $Z_i \subseteq A_i$ for every $i \in [-1 : \ell-1]$; and (2) $Z_{-1} = A_{-1}$.

Indistinguishable decomposition is helpful because when $\ell = L$, for every input assignment from $Z_{< L}$ to players $[-1 : L-1]$, the player -1 after L epochs (i.e., at the end of the protocol) sees the same transcript when player L receives different inputs from $R_{\geq L}$.⁷

⁷Due to our renaming, player -1 is the player 1 in the definition of the autoregressive model, who is supposed to output an answer at the end of the protocol.

In particular, it means for every $\tilde{z}_{< L} \in Z_{< L}$, the answer $L\text{-FuncComp}(\tilde{z}_{< L}, \tilde{z}_L)$ must be the same for every $\tilde{z}_L \in R_{\geq L}$. This is a strong constraint on the set $R_{\geq L}$ that we will analyze below.

First, we define $\mathcal{I}_{\ell-1} := \mathcal{I}_{\ell-1}(Z_{< L})$ as the set of all partial composition values $i_{\ell-1}$ for inputs from $Z_{< \ell}$.

Recall that by its definition, we have

$$L\text{-FuncComp}(\tilde{z}_{< L}, \tilde{z}_L) = i_L = \tilde{z}_L(\tilde{w}_{L-1}, i_{L-1}).$$

Fixing any $\tilde{z}_{< L} \in Z_{< L}$, this value is the same for all $\tilde{z}_L \in R_{\geq L}$. As $Z_{< L}$ is the product set $A_{-1} \times Z_0 \times \dots \times Z_{L-1}$ and $\tilde{z}_{< L} \in Z_{< L}$, variable $\tilde{w} = \tilde{z}_{-1}$ can take any value A_{-1} . So $\tilde{w}_{L-1}$ can take any value in $[n_{L-1}]$. Fixing any $\tilde{z}_{< L} \in Z_{< L}$ is equivalent to fixing any $\tilde{w}_{L-1} \in [n_{L-1}]$ and $i_{L-1} \in \mathcal{I}_{L-1}$, and the value of $\tilde{z}_L(\tilde{w}_{L-1}, i_{L-1})$ is fixed for every $\tilde{z}_L \in R_{\geq L}$. This in particular, means that

$$|R_{\geq L}| \leq \frac{|A_L|}{N_{L-1}^{|Z_{L-1}| \cdot n_{L-1}}}. \quad (1)$$

Here $\mathcal{I}_{L-1}$ is the set of partial composition values reachable from $Z_{< L}$. Fixing $i_{L-1} \in \mathcal{I}_{L-1}$ and $\tilde{w}_{L-1} \in [n_{L-1}]$ pins down one entry of the truth table of $\tilde{z}_L(\tilde{w}_{L-1}, i_{L-1}) : [n_{L-1}] \times [N_{L-1}] \rightarrow [N_L]$ (by indistinguishability constraint). On the numerator $|A_L| = N_{L-1}^{N_{L-1}} = N_{L-1}^{n_{L-1} \cdot N_{L-2}}$ is the total number of such truth tables, and each entry we pin down reduces the number of truth tables consistent with the indistinguishability constraint by a factor of N_{L-1} . Since $R_{\geq L}$ satisfies the indistinguishability constraint, it must be a subset of these truth tables.

Given the above (1), our goal now is (for every autoregressive communication protocol Π with L epochs and B message bits) to find an indistinguishable decomposition with the following two properties:

- **(Large remaining entropy)** The size of $R_{\geq L}$ is large; and
- **(Large cover)** The size of $\mathcal{I}_{L-1}(Z_{< L})$ is large.

We intentionally omit the precise meaning of “large” in the two items above; the idea is that these two conditions together would contradict (1), therefore showing Π does not compute $L\text{-FuncComp}$. Since Π is an arbitrary protocol, this proves the lower bound.

b) *Construct indistinguishable decomposition via induction.*: We construct the indistinguishable decomposition $R_{\geq \ell}$ and $Z_{< \ell}$ via an induction on ℓ . Suppose we have already obtained $R_{\geq \ell}$ and $Z_{< \ell}$, and let $\Lambda^{(\ell)} = \left\{ \Lambda_{\tilde{z}_{< \ell}}^{(\ell)} \right\}_{\tilde{z}_{< \ell} \in Z_{< \ell}}$ be the collection of transcripts from players $[\ell : L]$ to players $[-1 : \ell]$, when players $[\ell : L]$ take input assignments from $R_{\geq \ell}$ and players $[-1 : \ell-1]$ take input assignments from $Z_{< \ell}$. Indistinguishable decomposition ensures that $\Lambda^{(\ell)}$ is well-defined (e.g., it is the correct transcript for every $\tilde{z}_{\geq \ell} \in R_{\geq \ell}$).

To construct $R_{\geq \ell+1}$ and $Z_{< \ell+1}$, we proceed in two parts.

- **(Part 1)** First, we wish to select a set $Z_\ell \subseteq A_\ell$ and let $Z_{< \ell+1} = Z_{< \ell} \times Z_\ell$. To this end, we find a rectangular subset from $R_{\geq \ell}$, i.e., we find $S_{\geq \ell+1} \times Z_\ell \subseteq R_{\geq \ell}$ where $S_{\geq \ell+1} \subseteq A_L \times \dots \times A_{\ell+1}$ and $Z_\ell \subseteq A_\ell$, with the requirement that (1) $S_{\geq \ell+1}$ has large size, and (2) the size of $\mathcal{I}(Z_{< \ell+1})$ is large.
- **(Part 2)** Next, we want to distill a subset $R_{\geq \ell+1}$ from $S_{\geq \ell+1}$. This requires us to determine the transcripts from players $[\ell + 1 : L]$ to players $[-1 : \ell]$ in the first $\ell + 1$ epochs, when they receive input assignments from $R_{\geq \ell+1}$ and $Z_{< \ell+1}$. We further divide into three steps:
 - 1) **(Step 1)** We first determine transcripts to players $[-1 : \ell - 1]$ over the first ℓ epochs. We can simply use $\Lambda^{(\ell)}$ (i.e., transcripts from the inductive hypothesis). Up to this point, all assignments to $S_{\geq \ell+1}$ are indistinguishable to $Z_{< \ell+1}$.
 - 2) **(Step 2)** Next, we determine transcripts to players $[-1 : \ell - 1]$ in the $(\ell + 1)$ -th epoch. This is the key step of the whole proof. Our key insight is that Z_ℓ is indistinguishable to players $[-1 : \ell - 1]$ after ℓ epochs, when they take input from $Z_{< \ell}$. Hence, the $(\ell + 1)$ -th epoch transcripts to players $[-1 : \ell - 1]$ are independent of the choice of $z_\ell \in Z_\ell$. We can then use a greedy strategy to select transcripts that leak the least amount of information about $S_{\geq \ell+1}$, without consulting to the value of $z_\ell \in Z_\ell$.⁸ After this step, we are left with a large subset $T_{\geq \ell+1} \subseteq S_{\geq \ell+1}$ that is indistinguishable to $Z_{< \ell+1}$.
 - 3) **(Step 3)** Finally, we determine transcripts to players ℓ over the first $\ell + 1$ epochs, we simply use a greedy strategy to select transcripts that leak the least amount of information about $T_{\geq \ell+1}$, and we finally left with a large set $R_{\geq \ell+1} \subseteq T_{\geq \ell+1}$ that is indistinguishable to $Z_{< \ell+1}$.

D. Additional related work

a) *Representation power of Transformer*: There is a long line of work studying the representation power of Transformer [6], [13], [16], [19], [25], [29], [31], [36], [41], [49], [62], [63], [65]. Starting from the work of [23], [65], a line of work demonstrates the advantage of Transformer against recurrent architecture (RNNs, LSTMs and state space model) on a variety of tasks, such that parsing hierarchical structure [65], sparse averaging [49], in context learning [62] and copying [25].

Hahn [21] initiated the study on the limitation of the self-attention mechanism; the author proves that a Transformer with hard attention can not recognize parity

⁸This saving is crucial for our proof to work.

and dyck language. The lower bound does not extend to soft-attention since both parity and dyck language can be solved by two-layer Transformers. Since then, the literature either uses communication complexity to prove lower bound for one-layer Transformer [40], [49], [51], or puts Transformer into various computation models [37], [40], [51].

The CoT paradigm allows extra computation space for Transformer, and previous work [19], [29], [36], [41] demonstrated the power of CoT. [41] prove Transformer with arbitrary precise is Turing-complete, [36] and [29] prove that log-precision Transformer could simulate any polynomial time algorithm/polynomial-size circuit when the algorithm/circuit is given as input.

b) *Lower bounds against TC^0* : TC^0 is the class of constant-depth circuits consisting of AND, OR, NOT, and linear threshold gates.⁹ In the literature, the size of TC^0 circuits is often measured in terms of the number of wires (i.e., the total fan-in of all gates).

It is proven in [37] that constant-depth encoder-only transformer can be simulated by TC^0 circuits with roughly the same depth. Hence, strong lower bounds against TC^0 would imply lower bounds against a constant-depth encoder-only transformer. This simulation has at least a quadratic blow-up in terms of wires since it already requires at least an $\Omega(n^2)$ -wire TC^0 circuit to simulate a single attention layer. Unfortunately, the best known lower bounds against TC^0 circuits [12], [22], [24] are of the form n^{1+c-d} where $c > 1$ is a constant, which is not strong enough to imply any constant-depth encoder-only transformer lower bounds.¹⁰ Indeed, proving better lower bounds against TC^0 (say, $n^{1+1/d}$ wires against depth- d TC^0) is a major open question in complexity theory and would have breakthrough consequences [4], [11].

E. Discussion, limitation and future work

Our paper proves the first unconditional lower bound on the representational power of multi-layer Transformer, over a sequential composition task that could naturally arise from practice. Our results offer several interesting implications regarding the depth-size tradeoff inherent in Transformer architectures, the representational separation between encoders and decoders and the substantial impact of employing CoT. These theoretical insights align well with empirical observations, suggesting that the representational lens contributes meaningfully to understanding Transformer models.

⁹A linear threshold gate $G: \{0,1\}^n \rightarrow \{0,1\}$ outputs 1 on input $x \in \{0,1\}^n$ if $\sum_{i=1}^n w_i \cdot x_i \geq \theta$ and 0 otherwise, where $(w_1, \dots, w_n) \in \mathbb{R}^n$ is a weight vector and $\theta \in \mathbb{R}$ is a threshold.

¹⁰Also, the hard instance from [24], the parity function, can be solved by a decoder-only transformer with $\text{polylog}(n)$ model dimension. Here, the parity function takes n bits as input and outputs 1 if the number of ones in the input is odd, and 0 otherwise.

We primarily focus on theoretical analysis in an asymptotic regime, where the context length significantly surpasses the model dimension. While our theory qualitatively matches empirical observations, there exists a quantitative gap between our theory and the parameter settings typically utilized in practical models. Specifically, since state-of-the-art language models are already extremely large, our lower bound currently applies only to compositional tasks of astronomical size. Bridging this gap is an important avenue for future research.

Our work makes significant progress in the representation theory of neural networks by introducing new methods to analyze multi-layer Transformers, overcoming core limitations of prior approaches. We propose a novel communication model along with fundamentally new proof techniques, which differ substantially from all previous work: existing methods are not even applicable to analyzing depth-2 decoders. We believe our techniques can be further optimized, and we encourage future research to build upon and improve the bounds established in this work

a) *Future direction:* There are several interesting theoretical questions for future research. First of all, the lower bound in Theorem 1.1 only holds up to depth $L = o(\log \log(n))$ and the model dimension $Hdp \leq n^{-4^L}$ drops double exponentially with the number of layer L . We believe it is possible to improve it to a single-exponential decay (i.e., $n^{-O(L)}$) and strengthen the lower bound to log-depth Transformer. Currently, the $\log \log(n)$ factor comes from the fact that the function size growth ratio (i.e., $N_\ell/N_{\ell-1}$) is double exponentially decreasing in our construction, which is an ad-hoc limitation for our lower bound proof technique.

Open Question 1. *Prove the sequential function composition task is hard for log-depth Transformer. In particular, for any $L \leq \log(n)$, prove L -layer Transformer could not solve L -sequential function composition when $Hdp \leq n^{-O(L)}$.*

Our current construction inherently cannot surpass logarithmic depth, as the function size necessarily grows exponentially. A natural and intriguing question thus arises: Can we establish a polynomial-depth lower bound for Transformers? We propose a potential approach informed by two core observations underlying our current proof: (1) the information bottleneck induced by attention mechanisms, and (2) the autoregressive property inherent to decoders. We believe these insights can extend significantly further. Specifically, instead of explicitly providing the functions $f_1, \dots, f_L$ via truth tables in the prompt, one could succinctly encode these functions using compact circuit representations, thereby circumventing exponential description-size growth. Could such an encoding scheme enable proving a polynomial-depth

lower bound (e.g., $L \gtrsim n^{0.1}$), either unconditionally or based on standard computational complexity conjectures?

Open Question 2. *Prove a polynomial depth lower bound for Transformers, either unconditionally or conditioned on well-established computational complexity conjectures.*

II. PRELIMINARIES

A. Transformer

We formally describe the decoder-only Transformer architecture. Let L be the number of attention layers, H be the number of attention heads at each layer, p be the precision, d be the key, value and model dimension, n be the (input) prompt length. In order to carry out attention, it is common to assume the bit precision $p \geq \log(n)$.

An L -layer decoder-only Transformer is a sequence-to-sequence network, consists of alternating attention layer and MLP layer:

$$f_{\text{tran}} = f_{\text{mlp}}^{(L)} \circ f_{\text{attn}}^{(L)} \circ \dots \circ f_{\text{mlp}}^{(1)} \circ f_{\text{attn}}^{(1)}$$

Given an input sequence $x^{(0)} = (x_1^{(0)}, \dots, x_n^{(0)}) \in (\mathbb{R}^{dH})^n$, the Transformer inductively computes the output of the ℓ -th attention layer $y^{(\ell)} = (y_1^{(\ell)}, \dots, y_n^{(\ell)})$ and the output of the ℓ -th MLP layer $x^{(\ell)} = (x_1^{(\ell)}, \dots, x_n^{(\ell)})$. For layer $\ell = 1, 2, \dots, L$,

- **Attention layer $f_{\text{attn}}^{(\ell)}$:** For each attention head $h \in [H]$ and position $i \in [n]$, we have

$$y_i^{(\ell, h)} = \sum_{j \leq i} \alpha_{i,j}^{(\ell, h)} V^{(\ell, h)} x_j^{(\ell-1)} \in \mathbb{R}^d \quad (2)$$

where $\{\alpha_{i,j}^{(\ell, h)}\}_{j \leq i}$ is the attention probability of the h -th attention head, computed as

$$\begin{aligned} & \alpha_{i,j}^{(\ell, h)} \\ &= \frac{\exp((x_i^{(\ell-1)})^\top (Q^{(\ell, h)})^\top K^{(\ell, h)} x_j^{(\ell-1)})}{\sum_{j \leq i} \exp((x_i^{(\ell-1)})^\top (Q^{(\ell, h)})^\top K^{(\ell, h)} x_j^{(\ell-1)})} \\ & \in [0, 1] \end{aligned} \quad (3)$$

and $Q^{(\ell, h)}, K^{(\ell, h)}, V^{(\ell, h)} \in \mathbb{R}^{d \times dH}$ is the query, key and value matrix of the attention head.

Finally, the output of the ℓ -th attention layer is the concatenation of each head,

$$y_i^{(\ell)} = (y_i^{(\ell, 1)}, \dots, y_i^{(\ell, H)}) \in \mathbb{R}^{dH} \quad \forall i \in [n]$$

- **MLP layer $f_{\text{mlp}}^{(\ell)}$:** The output of the ℓ -th layer (and also the input to the $(\ell + 1)$ -th layer) is an arbitrary function $g^{(\ell)} : \mathbb{R}^{dH} \rightarrow \mathbb{R}^{dH}$ applied to each position:

$$x_i^{(\ell)} = g^{(\ell)}(y_i^{(\ell)}) \in \mathbb{R}^{dH}$$

Our formulation is general enough to capture detailed architecture choice such as the positional encoding, the residual connection, the layer normalization and the mixture-of-expert layers.

The only difference between Transformer encoder and decoder is that the encoder performs attention to all tokens (instead of previous token), so the summation in Eq. (2)(3) is over $j \in [n]$ (instead of $j \leq i$).

B. Sequential function composition

We use the following parameters throughout the paper.

$$K = (HdpL)^8 \cdot 8^{2L^2}, \quad m = K^{\sum_{\ell \in [0:L-1]} 8^{\ell+1}},$$

$$n_\ell = K^{4 \cdot 8^{L-\ell-1}} \quad \forall \ell \in [L-1]. \quad (4)$$

and

$$N_\ell = m \cdot \prod_{\ell' \in [\ell]} n_{\ell'} \quad \forall \ell \in [0:L-1]. \quad (5)$$

Definition II.1 (*L*-sequential function composition). A *L*-sequential function composition task $L\text{-FuncComp}(w, z_0, z_1, \dots, z_L)$ is described a sequence of functions $z_0, z_1, \dots, z_L$, where $z_0 \in [m]$ and $z_\ell : [N_{\ell-1}] \rightarrow [N_\ell]$ for $\ell \in [L]$ and a query $w = (w_1, \dots, w_{L-1}) \in [n_1] \times \dots \times [n_{L-1}]$, one computes

$$i_0 = z_0 \in [m], \quad i_1 = z_1(i_0) \in [N_0]$$

and one inductively computes, for each $\ell = 1, 2, \dots, L-1$:

$$i_2 = z_2(w_1, i_1) \in [N_1], \quad \dots,$$

$$i_{\ell+1} = z_{\ell+1}(w_\ell, i_\ell) \in [N_\ell] \quad (6)$$

The final output is taken as $L\text{-FuncComp}(w, z_0, z_1, \dots, z_L) = i_L$.

For Transformer to solve the *L*-sequential function composition, we assume the input prompt first describes *L* functions in the order of $z_{L-1}, \dots, z_0$, and then describes the query w . For simplicity, we assume each entry of z_ℓ ($\ell \in [0:L-1]$) is described using one token (so it takes $N_{\ell-1}$ tokens to describe z_ℓ); the query w is described in one token.

III. THE AUTOREGRESSIVE COMMUNICATION MODEL

In this section, we introduce a new communication model, which is a nice abstraction of the Transformer architecture for our purpose of proving lower bounds.

The Autoregressive Communication Model

a) Settings.: Let *L* be the number of epochs. There are *L* + 2 players; we call them players $[-1:L]$. They communicate in *L* epochs. Let

B be the message bits of a single token.

b) Input.: Player $i \in [-1:L]$ receives $m_{(i)}$ tokens z_i as input.

c) Communication.: The communication proceeds in *L* epochs. For $\ell \in [0:L]$, let $X_i^{(\ell)}$ be the message collected by player i ($i \in [-1:L]$) after the ℓ -th epoch of communication. Initially, when $\ell = 0$, $X_i^{(0)}$ is just the input of player i . For $\ell = 1, 2, \dots, L$, the ℓ -th epoch of communication proceeds as follows. For player $i \in [-1:L]$,

- The player i sends its information $X_i^{(\ell-1)}$ to all players $[i+1:L]$.
- The player $j \in [i+1:L]$, based on its own information $X_j^{(\ell-1)}$ and player i 's information $X_i^{(\ell-1)}$, it sends a message $\Pi_{j,i}^{(\ell)}$ to player i . The length of the message satisfies

$$|\Pi_{j,i}^{(\ell)}| = 2B \cdot m_{(i)}.$$

- Finally, the player i updates its collection of information as

$$X_i^{(\ell)} := X_i^{(\ell-1)} \cup \bigcup_{j>i} \Pi_{j,i}^{(\ell)}.$$

d) Output.: At the end of *L*-th round, the player -1 outputs a message based on its information $X_{-1}^{(L)}$.

Below, we state *L*-FuncComp in the autoregressive communication model formally.

L-FuncComp in the autoregressive communication model

e) Settings.: Let *L* be a parameter. Let K, m, n_ℓ, N_ℓ be defined according to (4) and (5). We set $B = Hdp$.

f) Input.: Player $i \in [L]$ receives z_i from Definition II.1 as an input, described in N_{i-1} tokens. Player 0 and player -1 receives z_0 and w as input, respectively, each described in 1 token.

g) Output.: At the end of *L*-th round, the player -1 needs to output $L\text{-FuncComp}(w, z_0, \dots, z_L)$.

A. Autoregressive communication lower bounds imply Transformer lower bounds

The following lemma shows that proving a lower bound for the above communication model immediately implies the desired lower bound against decoder-only Transformers.

Lemma III.1 (Reduction from Transformers to autoregressive communication). *If there is an L -layer decoder-only Transformer that solves the L -sequential function composition task, then there is a deterministic autoregressive communication protocol that solves L -FuncComp with L epochs and $B = Hdp$ message bits.*

Proof. Let Γ be an L -layer Transformer that solves the L -sequential function composition task. We use $K^{(\ell,h)}, Q^{(\ell,h)}, V^{(\ell,h)}$ to denote its key, query and value matrix, and $g^{(\ell)}$ to denote its MLP function. For $i \in [-1 : L]$, $E_i \subseteq [n]$ be the positions correspond to the input of player i , i.e.,

$$E_i = \begin{cases} \left[\sum_{j \in [i+1:L]} N_{j-1} + 1 : \sum_{j \in [i:L]} N_{j-1} \right] & i \in [L] \\ \left\{ \sum_{j \in [0:L-1]} N_j + 1 \right\} & i = 0 \\ \left\{ \sum_{j \in [0:L-1]} N_j + 2 \right\} & i = -1 \end{cases}$$

Consider the following autoregressive communication protocol for L -FuncComp with L epochs and $B = Hdp$. For epoch $\ell = 1, 2, \dots, L$, for each player i and for each $j > i$, define the transcript $\Pi_{j,i}^{(\ell)}$ from player j to player i as follow:

$$\begin{aligned} \Pi_{j,i}^{(\ell)} &= \left\{ \sum_{t \in E_j} \exp((x_r^{(\ell-1)})^\top (Q^{(\ell,h)})^\top K^{(\ell,h)} x_t^{(\ell-1)}) V^{(\ell,h)} x_t^{(\ell-1)} \right\}_{h \in [H], r \in E_i} \\ &\cup \left\{ \sum_{t \in E_j} \exp((x_r^{(\ell-1)})^\top (Q^{(\ell,h)})^\top K^{(\ell,h)} x_t^{(\ell-1)}) \right\}_{h \in [H], r \in E_i} \end{aligned}$$

The total length of $\Pi_{j,i}^{(\ell)}$ satisfies the requirement, i.e., we have

$$\begin{aligned} |\Pi_{j,i}^{(\ell)}| &= |E_i| \cdot H \cdot (dp + p) \leq |E_i| \cdot 2Hdp \\ &= \begin{cases} 2Hdp \cdot N_{i-1} & i \in [L] \\ 2Hdp & i \in \{0, -1\} \end{cases} \end{aligned}$$

We note that $\Pi_{j,i}^{(\ell)}$ depends only on the value of $\{x_r^{(\ell-1)}\}_{r \in E_i}$ and $\{x_t^{(\ell-1)}\}_{t \in E_j}$, hence, it remains to prove that they can be derived based on $\{X_r^{(\ell-1)}\}_{r \in E_j}$ and $\{X_t^{(\ell-1)}\}_{t \in E_i}$. To this end, we prove

Claim III.2. *For $\ell = 0, 1, \dots, L$, the player i ($i \in [-1 : L]$) knows the intermediate value $\{x_t^{(\ell)}\}_{t \in E_i}$ of the Transformer after ℓ -th epoch, i.e., $\{x_t^{(\ell)}\}_{t \in E_i}$ can be derived from $\{X_t^{(\ell)}\}_{t \in E_i}$.*

With Claim III.2 in hand, we can finish the proof since the player -1 knows the output embedding $x_n^{(L)}$ of the Transformer. $\square$

Proof of Lemma III.2. The inductive hypothesis is obviously true for $\ell = 0$. Suppose it continues to hold for $\ell - 1$, then for the ℓ -th epoch, the player i receives $\Pi_{j,i}^{(\ell)}$ for $j > i$, recall the output $\{y_r^{(\ell)}\}_{r \in E_i} = \{y_r^{(\ell,h)}\}_{r \in E_i, h \in [H]}$, and we have

$$\begin{aligned} y_r^{(\ell,h)} &= \sum_{t \leq r} \alpha_{r,t}^{(\ell,h)} V^{(\ell,h)} x_t^{(\ell-1)} \\ &= \frac{\sum_{j \geq i} \sum_{t \in E_j, t \leq i} \exp((x_r^{(\ell-1)})^\top (Q^{(\ell,h)})^\top K^{(\ell,h)} x_t^{(\ell-1)}) V^{(\ell,h)} x_t^{(\ell-1)}}{\sum_{j \geq i} \sum_{t \in E_j, t \leq i} \exp((x_r^{(\ell-1)})^\top (Q^{(\ell,h)})^\top K^{(\ell,h)} x_t^{(\ell-1)})} \end{aligned}$$

It can be obtained from the transcript $\{\Pi_{j,i}^{(\ell)}\}_{j > i}$ and $X_i^{(\ell-1)}$. Once the player i obtains $\{y_r^{(\ell)}\}_{r \in E_i}$, it can also obtain $\{x_r^{(\ell)}\}_{r \in E_i}$ since $x_r^{(\ell)} = g^{(\ell)}(y_r^{(\ell)})$ can be computed locally. This completes the proof. $\square$

IV. AUTOREGRESSIVE COMMUNICATION LOWER BOUND

Our main result (Theorem I.1) can be obtained from Lemma III.1 and the following lower bound for the communication problem.

Theorem IV.1. *There is no deterministic autoregressive communication protocol solving L -FuncComp with L epochs and $B = Hdp$ message bits.*

a) Notation.: For notational convenience, we use z_{-1} and w interchangeably to denote player -1 's input. In the following, we elaborate on several key definitions that will be crucial to our proof.

- (The transcript $\Pi_{j,i}^{(\ell)}$) For any $i \in [-1 : L-1]$, $j \in [i+1 : L]$, $\ell \in [L]$, recall $\Pi_{j,i}^{(\ell)}$ is the transcript sent from the player j to the player i at the ℓ -th epoch of communication. Its value is determined by the input of players $[i : L]$, i.e., $z_L, \dots, z_i$, and its value is independent of the choice of $z_{i-1}, \dots, z_0, w$.¹¹ For any fixed value $\tilde{z}_L \in [N_{L-1}]^{N_{L-1}}, \dots, \tilde{z}_i \in [N_{i-1}]^{N_{i-1}}$, let $\Pi_{j,i}^{(\ell)}(\tilde{z}_L, \dots, \tilde{z}_i)$ be the transcript when the players t receives input $z_t = \tilde{z}_t$ ($t \in [i : L]$).
- (The partial composition value $i_\ell(\tilde{w}, \tilde{z}_0, \dots, \tilde{z}_\ell)$) For any $\ell \in [0 : L]$, the value of i_ℓ is determined by $w, z_0, \dots, z_\ell$. We write $i_\ell(\tilde{w}, \tilde{z}_0, \dots, \tilde{z}_\ell)$ to denote the value of i_ℓ when $w = \tilde{w}, z_0 = \tilde{z}_0, \dots, z_\ell = \tilde{z}_\ell$.
- b) Parameters.:* We use the following parameters

$$\begin{aligned} x_\ell &= K^{8^{L-\ell-1}} (\forall \ell \in [0 : L-1]), \\ A_\ell &= \left[N_{\ell-1}^{N_{\ell-1}} \right] \quad (\forall \ell \in [L]) \end{aligned} \quad (7)$$

and

$$\begin{aligned} \Delta_\ell &= 2^{4\sqrt{K}(x_0 \dots x_{\ell-2}) \cdot (n_1 \dots n_{L-1})} \quad (\forall \ell \in [2 : L]), \\ \Theta_\ell &= 8^{-L\ell} (x_0 \dots x_\ell) \cdot (n_1 \dots n_{\ell-1}) \quad (\forall \ell \in [L-1]). \end{aligned} \quad (8)$$

¹¹The transcript also depends on the underlying autoregressive communication protocol Π , which will be clear from the context.

For notational convenience, we also set $A_{-1} = \prod_{i=1}^{L-1} [n_i]$ and $A_0 = [m]$. Note that with our convention of denoting w by z_{-1} , player i takes an input from A_i for every $i \in [-1 : L]$.¹²

The meaning of parameters will be clearer after we state our main technical centerpiece Lemma IV.5.

c) *Indistinguishable decomposition.*: Our key idea is the following concept of *indistinguishable decomposition*, which is two sets $R_{\geq \ell}$ and $Z_{< \ell}$, where $R_{\geq \ell}$ is a set of input assignments to players $[\ell : L]$ and $Z_{< \ell}$ is a set of input assignments to players $[-1 : \ell - 1]$, such that for every possible inputs $z_{< \ell} \in Z_{< \ell}$, all assignments to $R_{\geq \ell}$ are indistinguishable to players $[-1 : \ell - 1]$ on inputs $z_{< \ell}$ after ℓ epochs (because they lead to the same transcripts).

Formally, we define:

Definition IV.2 (Indistinguishable decomposition). Let $\ell \in [2 : L]$,

$$R_{\geq \ell} \subseteq A_L \times A_{L-1} \times \cdots \times A_\ell$$

and

$$Z_{< \ell} = Z_{-1} \times \cdots \times Z_{\ell-1} \subseteq A_{-1} \times \cdots \times A_{\ell-1}$$

where

$$Z_{-1} = A_{-1}, Z_0 \subseteq A_0, \dots, Z_{\ell-1} \subseteq A_{\ell-1}.$$

We say $R_{\geq \ell}$ and $Z_{< \ell}$ is an *indistinguishable decomposition*, if for every $\tilde{z}_{< \ell} \in Z_{< \ell}$, and for every $\tilde{\alpha}_{\geq \ell}, \tilde{\beta}_{\geq \ell} \in R_{\geq \ell}$, it satisfies:

$$\Pi_{j,i}^{(\ell')}(\tilde{z}_{< \ell}, \tilde{\alpha}_{\geq \ell}) = \Pi_{j,i}^{(\ell')}(\tilde{z}_{< \ell}, \tilde{\beta}_{\geq \ell})$$

for every $j \in [\ell : L]$, $i \in [-1 : \ell - 1]$, and $\ell' \in [\ell]$.

Indistinguishable configuration is helpful because when $\ell = L$, for every input assignment from $Z_{< L}$ to players $[-1 : L - 1]$, the player -1 after L epochs (i.e., at the end of the protocol) sees the same transcript when player L receives inputs from $R_{\geq L}$. In particular, it means for every $\tilde{z}_{< L} \in Z_{< L}$, the answer $L\text{-FuncComp}(\tilde{z}_{< L}, \tilde{z}_L)$ must be the same for every $\tilde{z}_L \in R_{\geq L}$. This is a strong requirement and we will carefully define properties of $R_{\geq \ell}$ and $Z_{< \ell}$ such that this requirement would lead to contradiction, and therefore obtaining our lower bound.

For a subset $Z_{< \ell}$, we define $\mathcal{I}_{\ell-1}$ as

$$\mathcal{I}_{\ell-1}(Z_{< \ell}) := \{\tilde{i}_{\ell-1} : \tilde{i}_{\ell-1} = i_{\ell-1}(\tilde{z}_{-1}, \tilde{z}_0, \dots, \tilde{z}_{\ell-1}) \text{ for some } (\tilde{z}_{-1}, \tilde{z}_0, \dots, \tilde{z}_{\ell-1}) \in Z_{< \ell}\}.$$

That is, $\mathcal{I}_{\ell-1}(Z_{< \ell})$ is the set of all partial composition values for inputs from $Z_{< \ell}$.

¹²Here, we can interpret an element from $[N_{\ell-1}^{N_{\ell-1}-1}]$ as a function from $[N_{\ell-1}] \rightarrow [N_{\ell-1}]$ via a natural bijection.

The following lemma shows that the desired lower bound follows from a good enough indistinguishable configuration for $\ell = L$.

Lemma IV.3. Let Π be an L -epoch Hdp message bits autoregressive communication protocol. If there is an indistinguishable decomposition $R_{\geq L}$ and $Z_{< L}$ such that:

- 1) (**Large remaining entropy**) $|R_{\geq L}| \geq |A_L|/\Delta_L$.
- 2) (**Large cover**) $|\mathcal{I}_{L-1}(Z_{< L})| \geq \Theta_{L-1}$.

Then Π does not solve $L\text{-FuncComp}$.

In the next subsection, we will show the existence of the required indistinguishable decomposition from Lemma IV.3 via an induction, which finishes the proof of Theorem IV.1.

Lemma IV.4. For every L -epoch Hdp message bits autoregressive communication protocol Π , there is an indistinguishable decomposition $R_{\geq L}$ and $Z_{< L}$ satisfying the requirements of Lemma IV.3.

We finish this subsection by proving Lemma IV.3.

Proof of Lemma IV.3. First, by the large remain entropy property and our choice of parameters, we have

$$\begin{aligned} |R_{\geq L}| &\geq |A_L|/\Delta_L = |A_L| \cdot 2^{-4\sqrt{K}x_0 \cdots x_{L-2} \cdot n_1 \cdots n_{L-1}} \\ &> |A_L| \cdot 2^{-8^{-L^2}(x_0 \cdots x_{L-1})(n_1 \cdots n_{L-1})} \\ &\geq |A_L| \cdot 2^{-n_{L-1}\Theta_{L-1}} \\ &\geq |A_L| \cdot 2^{-n_{L-1}|\mathcal{I}_{L-1}(Z_{< L})|} \\ &\geq \frac{|A_L|}{(N_{L-1})^{n_{L-1}|\mathcal{I}_{L-1}(Z_{< L})|}}. \end{aligned}$$

Here the second step follows from the definition of Δ_L (see Eq. (8)), the third and fourth step follow from the choice of parameters (see Eq. (4)(7)), the fifth step follows from $|\mathcal{I}_{\ell-1}(Z_{< L})| \geq \Theta_{\ell-1}$, i.e., the large cover property.

Hence, there exists $\tilde{i}_{L-1} \in \mathcal{I}_{L-1}(Z_{< L})$ and $\tilde{w}_{L-1} \in [n_{L-1}]$, such that, there exists $z'_L, z''_L \in R_{\geq L}$ satisfying $z'_L(\tilde{w}_{L-1}, \tilde{i}_{L-1}) \neq z''_L(\tilde{w}_{L-1}, \tilde{i}_{L-1})$. By definition, there exists $\tilde{z}_{< L} \in Z_{< L}$, such that (1) $\tilde{i}_{L-1} = i_{L-1}(\tilde{z}_{< L})$ and (2) $\tilde{z}_{-1, L-1} = \tilde{w}_{L-1}$. By the definition of $R_{\geq L}$, the messages sent from player L to players $[-1 : L - 1]$ are the same under z'_L, z''_L , when the ℓ -th player ($\ell \in [-1 : L - 1]$) receives $\tilde{z}_\ell$ as input. Hence, the player -1 can not distinguish between z'_L and z''_L , and therefore, it could not be correct on answering both $i'_L = z'_L(\tilde{w}_{L-1}, \tilde{i}_{L-1})$ and $i''_L = z''_L(\tilde{w}_{L-1}, \tilde{i}_{L-1})$, since $z'_L(\tilde{w}_{L-1}, \tilde{i}_{L-1}) \neq z''_L(\tilde{w}_{L-1}, \tilde{i}_{L-1})$. $\square$

A. Constructing Indistinguishable Decompositions via Induction

The rest of this section is devoted to prove Lemma IV.4, we will indeed prove it via an induction specified

in the Lemma IV.5, whose $\ell = L$ case is exactly Lemma IV.4.

Lemma IV.5 (Main Lemma). *For any $\ell \in [2 : L]$,*

- *We have a pair of sets $(R_{\geq \ell}, Z_{< \ell})$, where $R_{\geq \ell} \subseteq A_L \times A_{L-1} \times \dots \times A_\ell$, $Z_{< \ell} = Z_{-1} \times Z_0 \times \dots \times Z_{\ell-1}$, with $Z_{-1} = [n_1 \dots n_{L-1}]$, $Z_0 \subseteq A_0$, $Z_1 \subseteq A_1, \dots, Z_{\ell-1} \subseteq A_{\ell-1}$ and they have size $|Z_0| = x_0, |Z_1| = x_1, \dots, |Z_{\ell-1}| = x_{\ell-1}$;*
- *We can fix the transcript from players $[\ell : L]$ to $[-1 : \ell - 1]$ at the first ℓ epochs, when the players $[-1 : \ell - 1]$ take input from $Z_{< \ell}$. i.e.,*

$$\Lambda^{(\ell)} := \left(\Lambda_{j,i}^{(\ell, \ell')} \right)_{j \in [\ell : L], i \in [-1 : \ell - 1], \ell' \in [\ell]}$$

where

$$\Lambda_{j,i}^{(\ell, \ell')} := \left(\Lambda_{j,i}^{(\ell, \ell')}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \right)_{\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i}$$

and

$$\Lambda_{j,i}^{(\ell, \ell')}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \in \text{domain}(\Pi_{j,i}^{(\ell')})$$

such that we have the following guarantees:

- **(Consistency)** $\Lambda^{(\ell)}$ is the first ℓ -epoch transcript from players $[\ell : L]$ to $[-1 : \ell - 1]$, when they take input from $R_{\geq \ell}$ and $Z_{< \ell}$, i.e.,

$$\begin{aligned} \Pi_{j,i}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_i) &= \Lambda_{j,i}^{(\ell, \ell')}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \\ \forall j &\in [\ell : L], i \in [-1 : \ell - 1], \ell' \in [\ell], \\ \tilde{z}_{\geq L} &\in R_{\geq L}, \tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i, \end{aligned}$$

- **(Large remaining entropy)** The size of $R_{\geq \ell}$ is large, i.e.,

$$|R_{\geq \ell}| \geq |A_L| \dots |A_\ell| / \Delta_\ell. \quad (9)$$

- **(Large cover)** The total number of possible $i_{\ell-1}$ under $Z_{-1}, Z_0, Z_1, \dots, Z_{\ell-1}$ is large, i.e.,

$$\begin{aligned} \mathcal{I}_{\ell-1} &:= \{\tilde{i}_{\ell-1} : \tilde{i}_{\ell-1} = i_{\ell-1}(\tilde{w}, \tilde{z}_0, \dots, \tilde{z}_{\ell-1}) \\ &\text{for some } \tilde{w} \in Z_{-1}, \tilde{z}_0 \in Z_0, \dots, \tilde{z}_{\ell-1} \in Z_{\ell-1}\} \end{aligned}$$

and its size satisfies

$$|\mathcal{I}_{\ell-1}| \geq \Theta_{\ell-1}. \quad (10)$$

B. The Initial step

We first prove the correctness of Lemma IV.5 for $\ell = 2$.

1) *Step 1: Choosing Z_0, Z_1 :* We take $Z_0 = [x_0]$ and our first step is to select the set $Z_1 \subseteq [N_0]$. To this end, consider all possible first epoch messages from the player 1 to the player -1 , i.e.,

$$\begin{aligned} \Psi_{1,-1}^{(1)} &= \left(\Psi_{1,-1}^{(1)}(\tilde{z}_{-1}) \right)_{\tilde{z}_{-1} \in Z_{-1}} \\ \text{where } \Psi_{1,-1}^{(1)}(\tilde{z}_{-1}) &\in \{0, 1\}^{2Hdp}. \end{aligned}$$

The total number of possible $\Psi_{1,-1}^{(1)}$ is $2^{2Hdp \cdot |Z_{-1}|} = 2^{2Hdp(n_1 \dots n_{L-1})}$, and therefore, there exists one $\tilde{\Psi}_{1,-1}^{(1)} \in \{0, 1\}^{2Hdp \cdot (n_1 \dots n_{L-1})}$, such that

$$\begin{aligned} S &:= \{\tilde{z}_1 \in A_1 : \Pi_{1,-1}^{(1)}(\tilde{z}_1, \tilde{z}_{-1}) = \tilde{\Psi}_{1,-1}^{(1)}(\tilde{z}_{-1}) \\ &\quad \forall \tilde{z}_{-1} \in Z_{-1}\} \subseteq A_1 \end{aligned}$$

and

$$|S| \geq |A_1| \cdot 2^{-2Hdp \cdot (n_1 \dots n_{L-1})}. \quad (11)$$

Note the first epoch message depends only on $\tilde{z}_1$ and $\tilde{z}_{-1}$ so we can write it as $\Pi_{1,-1}^{(1)}(\tilde{z}_1, \tilde{z}_{-1})$.

The proof of the following Lemma can be found at Section IV-D.

Lemma IV.6. *There exists a subset $Z_1 \subseteq S$ with size $|Z_1| = x_1$, such that it satisfies*

$$|\{\tilde{z}_1(i_0) : \tilde{z}_1 \in Z_1, i_0 \in Z_0\}| \geq 8^{-L} x_0 x_1 = \Theta_1. \quad (12)$$

We take the subset Z_1 in Lemma IV.6 and it remains to fix the transcripts from players $j \in [2 : L - 1]$ to players $i = -1, 0, 1$ at the first two epochs.

2) *Step 2.1: Fixing the transcript to player -1 :* We first fix the transcript to player -1 . For the first epoch, we need to fix $\Lambda_{j,-1}^{(2,1)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1})$ for every $\tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}$. We note that, the first epoch message from player j to player -1 depends only on $\tilde{z}_{-1}$ and player j 's input, but not on $\tilde{z}_1, \tilde{z}_0$, hence it suffices to find some

$$\begin{aligned} \Phi_{j,-1}^{(1)} &= \left(\Phi_{j,-1}^{(1)}(\tilde{z}_{-1}) \right)_{\tilde{z}_{-1} \in Z_{-1}} \\ \text{where } \Phi_{j,-1}^{(1)}(\tilde{z}_{-1}) &\in \{0, 1\}^{2Hdp}. \end{aligned}$$

and set

$$\begin{aligned} \Lambda_{j,-1}^{(2,1)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1}) &= \Phi_{j,-1}^{(1)}(\tilde{z}_{-1}) \\ \forall \tilde{z}_1 &\in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1} \end{aligned}$$

The total number of such transcripts are at most $2^{2Hdp \cdot |Z_{-1}|} = 2^{2Hdp \cdot (n_1 \dots n_{L-1})}$. Hence, we can choose $\{\Lambda_{j,-1}^{(2,1)}\}_{j \in [2:L]}$, such that the set of consistent $\tilde{z}_L, \dots, \tilde{z}_2$,

$$C_1 := \left\{ \begin{aligned} &(\tilde{z}_L, \dots, \tilde{z}_2) \in A_L \times \dots \times A_2 : \\ &\Pi_{j,-1}^{(1)}(\tilde{z}_L, \dots, \tilde{z}_0, \tilde{z}_{-1}) = \Lambda_{j,-1}^{(2,1)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1}) \\ &\forall \tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}, j \in [2 : L] \end{aligned} \right\}.$$

satisfies

$$|C_1| \geq |A_L| \dots |A_2| \cdot 2^{-2Hdp \cdot (n_1 \dots n_{L-1}) \cdot L}.$$

For the second epoch, we need to fix $\Lambda_{j,-1}^{(2,2)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1})$ for every $\tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}$. The transcript from player $j \in [2 : L]$ to player -1 depends only on the information state $X_{-1}^{(1)}$ and $X_j^{(1)}$, which are independent of the choice of $\tilde{z}_1 \in Z_1$.

This is because, the only message in $X_{-1}^{(1)}$ and $X_j^{(1)}$ that depends on z_1 is the first epoch message from player 1 to -1 , which equals to $\Psi_{1,-1}^{(1)}(\tilde{z}_{-1})$ (see Eq. (11) and Lemma IV.6) and it is the same for every $\tilde{z}_1 \in Z_1$. Hence it suffices to find some

$$\Phi_{j,-1}^{(2)} := \left(\Phi_{j,-1}^{(2)}(\tilde{z}_0, \tilde{z}_{-1}) \right)_{\tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}}$$

and set

$$\Lambda_{j,-1}^{(2,2)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1}) = \Phi_{j,-1}^{(2)}(\tilde{z}_0, \tilde{z}_{-1}) \\ \forall \tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}$$

The total number of such transcripts are at most $2^{2Hdp \cdot x_0 \cdot (n_1 \dots n_{L-1})}$. Hence, we can properly choose $\{\Lambda_{j,-1}^{(2,2)}\}_{j \in [2:L-1]}$, such that the set of consistent $(z_L, \dots, z_2)$

$$C_2 := \left\{ \begin{array}{l} (\tilde{z}_L, \dots, \tilde{z}_2) \in C_1 : \\ \Pi_{j,-1}^{(2)}(\tilde{z}_L, \dots, \tilde{z}_0, \tilde{z}_{-1}) = \Lambda_{j,-1}^{(2,2)}(\tilde{z}_1, \tilde{z}_0, \tilde{z}_{-1}) \\ \forall \tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in Z_{-1}, j \in [2:L] \end{array} \right\}.$$

satisfies

$$|C_2| = |C_1| \cdot 2^{-2Hdp \cdot x_0 \cdot (n_1 \dots n_{L-1}) \cdot L} \\ \geq |A_L| \dots |A_2| \cdot 2^{-4HdpL \cdot x_0 \cdot (n_1 \dots n_{L-1})}$$

3) *Step 2.2: Fixing the transcript to player 0:* We then fix the transcript to player 0. The total number of transcripts $\{\Lambda_{j,0}^{(2,\ell')}\}_{j \in [2:L], \ell' \in [2]}$ of the first two epochs is at most $2^{2Hdp \cdot x_0 \cdot 2L}$. We can fix its value so that the set of consistent $(z_L, \dots, z_2)$

$$C_3 := \left\{ \begin{array}{l} (\tilde{z}_L, \dots, \tilde{z}_2) \in C_2 : \\ \Pi_{j,0}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_0) = \Lambda_{j,0}^{(2,\ell')}(\tilde{z}_1, \tilde{z}_0) \\ \forall \tilde{z}_1 \in Z_1, \tilde{z}_0 \in Z_0, j \in [2:L], \ell' \in [2] \end{array} \right\}.$$

satisfies

$$|C_3| \geq |C_2| \cdot 2^{-2Hdp \cdot x_0 \cdot 2L} \\ \geq |A_L| \dots |A_2| \cdot 2^{-6HdpL \cdot x_0 \cdot (n_1 \dots n_{L-1})}.$$

4) *Step 2.3: Fixing the transcript to player 1:* Finally, we fix the transcript to player 1. The total number of transcripts $\{\Lambda_{j,1}^{(2,\ell')}\}_{j \in [2:L], \ell' \in [2]}$ of the first two epochs is at most $2^{2Hdp \cdot x_1 \cdot 2L}$, and we can fix the value so that the set of consistent $(z_L, \dots, z_2)$

$$C_4 := \left\{ \begin{array}{l} (\tilde{z}_L, \dots, \tilde{z}_2) \in C_3 : \\ \Pi_{j,1}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_1) = \Lambda_{j,1}^{(2,\ell')}(\tilde{z}_1) \\ \forall \tilde{z}_1 \in Z_1, j \in [2:L], \ell' \in [2] \end{array} \right\}.$$

and we have

$$C_4 \geq C_3 \cdot 2^{-2Hdp \cdot x_1 \cdot 2L} \\ \geq |A_L| \dots |A_2| \cdot 2^{-8HdpL \cdot x_0 \cdot (n_1 \dots n_{L-1})} \\ \geq |A_L| \dots |A_2| \cdot 2^{-4\sqrt{K}x_0 \cdot (n_1 \dots n_{L-1})} \\ = |A_L| \dots |A_2| / \Delta_2 \quad (13)$$

Here the second step follows from the choice of parameters (see Eq. (4)(7)) and the last step follows from the definition of Δ_2 (see Eq. (8)).

Combining Lemma IV.6 and Eq. (13), we conclude the proof for the case $\ell = 2$.

C. Inductive step

Suppose Lemma IV.5 holds up to $\ell \in [2:L-1]$, we prove it continues to hold for $\ell+1$. This proceeds in a few steps:

- We first select the set Z_ℓ . To this end, we find a rectangular subset from $R_{\geq \ell}$, i.e., we find $S_{\geq \ell+1} \times Z_\ell \subseteq R_{\geq \ell}$ where $S_{\geq \ell+1} \subseteq A_L \times \dots \times A_{\ell+1}$ and $Z_\ell \subseteq A_\ell$, with the requirement that (1) $S_{\geq \ell+1}$ has large size, and (2) the size of $\mathcal{I}(Z_{<\ell+1})$ is large. see Section IV-C1 for details.
- We then fix the transcripts to players $[-1:\ell-1]$ in the first ℓ epochs, for which we simply use the transcript $\Lambda^{(\ell)}$ from the inductive hypothesis; see Section IV-C2 for details.
- Next, we fix the transcripts to players $[-1:\ell-1]$ for the $(\ell+1)$ -th epoch. This is the key step of our proof. Our key insight is that Z_ℓ is indistinguishable to players $[-1:\ell-1]$ after ℓ epochs, when they take input from $Z_{<\ell}$. Hence, the $(\ell+1)$ -th epoch transcripts to players $[-1:\ell-1]$ are independent of the choice of $z_\ell \in Z_\ell$. We can then use a greedy strategy to select transcripts that leak the least amount of information, without consulting to the value of $z_\ell \in Z_\ell$; see Section IV-C3 for details.
- Finally, we fix the transcripts to players ℓ . We use a greedy strategy and select the transcript that leaks the least amount of information; see Section IV-C4 for details.

Our key insight is that Z_ℓ is indistinguishable to players $[-1:\ell-1]$ after ℓ epochs, when they take input from $Z_{<\ell}$. Hence, the $(\ell+1)$ -th epoch transcripts to players $[-1:\ell-1]$ are independent of the choice of $z_\ell \in Z_\ell$. We can then use a greedy strategy to select transcripts that leak the least amount of information about $S_{\geq \ell+1}$, without consulting to the value of $z_\ell \in Z_\ell$.

1) *Step 1: Choosing the set Z_ℓ :* Recall the size of $R_{\geq \ell}$ satisfies

$$|R_{\geq \ell}| \geq |A_L| \times \dots \times |A_\ell| / \Delta_\ell. \quad (14)$$

We would like to select a rectangular subset from $R_{\geq \ell}$. The proof of Lemma IV.7 is deferred to Section IV-D.

Lemma IV.7. *There exists a subset $S^{(\ell)} \subseteq R_{\geq \ell}$ such that*

- $S^{(\ell)} = S_1^{(\ell)} \times S_2^{(\ell)}$, where $S_1^{(\ell)} \subseteq A_L \times \dots \times A_{\ell+1}$, $S_2^{(\ell)} \subseteq A_\ell$, with size

$$|S_1^{(\ell)}| \geq |A_L| \dots |A_{\ell+1}| / \Delta_\ell^{2x_\ell} \quad \text{and} \quad |S_2^{(\ell)}| = x_\ell.$$

• We have

$$\begin{aligned} & |\{i_\ell : i_\ell = \tilde{z}_\ell(\tilde{w}_{\ell-1}, \tilde{i}_{\ell-1}) \text{ for some } \tilde{w}_{\ell-1} \in [n_{\ell-1}], \\ & \tilde{i}_{\ell-1} \in \mathcal{I}_{\ell-1}, \tilde{z}_\ell \in S_2^{(\ell)}\}| \\ & \geq \Theta_\ell \end{aligned}$$

With Lemma IV.7 in hand, we take $Z_\ell = S_2^{(\ell)}$ and $S_{\geq \ell+1} = S_1^{(\ell)}$.

Next, we are going to fix the transcript $\Lambda^{(\ell+1)}$. Recall we need to fix all transcripts from players $[\ell+1 : L]$ to players $[-1 : \ell]$ in the first $\ell+1$ epochs, when players $[-1 : \ell]$ receive input from $Z_{\leq \ell} = Z_{-1} \times Z_0 \times \dots \times Z_\ell$. We proceed in a few steps.

2) *Step 2.1: Fixing the transcript to players $[-1 : \ell-1]$ in the first ℓ epochs:* First, we fix the transcript from players $j \in [\ell+1 : L]$ to players $i \in [-1 : \ell-1]$ in the first ℓ epochs. We simply use $\Lambda^{(\ell)}$, that is, for any $\tilde{z}_\ell \in Z_\ell, \dots, \tilde{z}_i \in Z_i$,

$$\begin{aligned} \Lambda_{j,i}^{(\ell+1,\ell')}(\tilde{z}_\ell, \dots, \tilde{z}_i) &= \Lambda_{j,i}^{(\ell,\ell')}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i). \\ \forall j \in [\ell+1 : L], i \in [-1 : \ell-1], \ell' \in [\ell] \end{aligned} \quad (15)$$

We claim that $S_{\geq \ell+1} \subseteq A_L \times \dots \times A_{\ell+1}$ is consistent with $\Lambda^{(\ell+1)}$ up to this point. Formally, we have

Lemma IV.8. *The set $S_{\geq \ell+1}$ is consistent with $\{\Lambda_{j,i}^{(\ell,\ell')}\}_{j \in [\ell+1:L], i \in [-1:\ell-1], \ell' \in [\ell]}$. Formally, for any $\tilde{z}_{\geq \ell+1} \in S_{\geq \ell+1}$ and $\tilde{z}_{< \ell+1} \in Z_{< \ell+1}$, one has*

$$\Pi_{j,i}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_i) = \Lambda_{j,i}^{(\ell+1,\ell')}(\tilde{z}_\ell, \dots, \tilde{z}_i).$$

for any $j \in [\ell+1 : L], i \in [-1 : \ell-1], \ell' \in [\ell]$.

Proof. By Lemma IV.7 and our choice of Z_ℓ , we have $(\tilde{z}_L, \dots, \tilde{z}_\ell) \in S_{\geq \ell+1} \times Z_\ell \subseteq R_{\geq \ell}$, and therefore

$$\begin{aligned} \Pi_{j,i}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_i) &= \Lambda_{j,i}^{(\ell,\ell')}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \\ &= \Lambda_{j,i}^{(\ell+1,\ell')}(\tilde{z}_\ell, \tilde{z}_{\ell-1}, \dots, \tilde{z}_i). \end{aligned}$$

where the first step follows from the definition of $R_{\geq \ell}$, the second step follows from Eq. (15). $\square$

3) *Step 2.2: Fixing the transcript to players $[-1 : \ell-1]$ at the $(\ell+1)$ -th epoch:* Next, we fix the transcript from players $j \in [\ell+1 : L]$ to players $i \in [-1 : \ell-1]$ at the $(\ell+1)$ -th epoch. Our key insight is that Z_ℓ is indistinguishable to players $[-1 : \ell-1]$ when they take input from $Z_{\leq \ell-1}$, hence their transcripts are independent of the choice of $z_\ell \in Z_\ell$. To this end, consider

$$\Phi^{(\ell+1)} = \left(\Phi_{j,i}^{(\ell+1)} \right)_{j \in [\ell+1:L], i \in [-1:\ell-1]}$$

where

$$\begin{aligned} \Phi_{j,i}^{(\ell+1)} &= \left(\Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \right)_{\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i} \\ \text{and } \Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) &\in \text{domain}(\Pi_{j,i}^{(\ell+1)}) \end{aligned}$$

Comparing $\Lambda_{j,i}^{(\ell+1,\ell+1)}$ with $\Phi_{j,i}^{(\ell+1)}$, we note that $\Phi_{j,i}^{(\ell+1)}$ does not have dependence on $\tilde{z}_\ell \in Z_\ell$.

For any $\Phi^{(\ell+1)}$, define

$$S(\Phi^{(\ell+1)}) := \left\{ \begin{array}{l} (\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in S_{\geq \ell+1} : \text{such that} \\ \Pi_{j,i}^{(\ell+1)}(\tilde{z}_L, \dots, \tilde{z}_i) = \Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \\ \forall \tilde{z}_\ell \in Z_\ell, \dots, \tilde{z}_i \in Z_i, \\ j \in [\ell+1 : L], i \in [-1 : \ell-1] \end{array} \right\} \quad (16)$$

In words, $S(\Phi^{(\ell+1)})$ include all $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in S_{\geq \ell+1}$ that are consistent with the transcript $\Phi^{(\ell+1)}$. Our key observation is

Lemma IV.9. *We have*

$$\bigcup_{\Phi^{(\ell+1)}} S(\Phi^{(\ell+1)}) = S_{\geq \ell+1}.$$

Proof. It suffices to prove that, for any $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in S_{\geq \ell+1}$, $\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i$, $j \in [\ell+1 : L], i \in [-1 : \ell-1]$, the transcript $\Pi_{j,i}^{(\ell+1)}(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}, \tilde{z}_\ell, \tilde{z}_{\ell-1}, \dots, \tilde{z}_i)$ is the same for every $z_\ell \in Z_\ell$.

To prove this, note that the transcript is determined by the information state $X_j^{(\ell)}$ and $X_i^{(\ell)}$. It is clear that $X_j^{(\ell)}$ does not change with the choice of $z_\ell \in Z_\ell$ since $j > \ell$. It remains to prove that $X_i^{(\ell)}$ also does not change with $z_\ell \in Z_\ell$. To this end, we prove that all information states $\{X_r^{(\ell')}\}_{r \in [i:\ell-1], \ell' \in [\ell]}$ does not change with z_ℓ .

Recall we have fixed the value of $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in S_{\geq \ell+1}$ and $\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i$. We prove by induction on $r = \ell-1, \dots, i$. When $r = \ell-1$, the information state $X_{\ell-1}^{(\ell')}$ is determined by $\tilde{z}_{\ell-1}$ and $\Pi_{t,\ell-1}^{(\ell'')}(z_L, \dots, z_{\ell+1}, z_\ell, \tilde{z}_{\ell-1})$ ($t \in [\ell : L], \ell'' \in [\ell']$), since $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}, z_\ell) \in R_{\geq \ell}$ for every $z_\ell \in Z_\ell$, we have that $\Pi_{t,\ell-1}^{(\ell'')}(z_L, \dots, z_{\ell+1}, z_\ell, \tilde{z}_{\ell-1}) = \Lambda_{t,\ell-1}^{(\ell,\ell'')}(z_{\ell-1})$, which is the same for every $z_\ell \in Z_\ell$. This finishes the proof of the base case.

Now suppose the induction continues to hold for $r+1$, for r , the information state $X_r^{(\ell')}$ is determined by $\tilde{z}_r$, $\Pi_{t,r}^{(\ell'')}(z_L, \dots, z_{\ell+1}, z_\ell, \tilde{z}_{\ell-1}, \dots, \tilde{z}_r)$ ($t \in [r+1 : L], \ell'' \in [\ell']$). For $t \in [\ell : L]$, since $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}, z_\ell) \in R_{\geq \ell}$, we have that $\Pi_{t,r}^{(\ell'')}(z_L, \dots, z_{\ell+1}, z_\ell, \tilde{z}_{\ell-1}, \dots, \tilde{z}_r) = \Lambda_{t,r}^{(\ell,\ell'')}(z_{\ell-1}, \dots, \tilde{z}_r)$, which are the same for every $z_\ell \in Z_\ell$. For $t \in [r+1 : \ell-1]$, we have proved $X_t^{(\ell')}$ are the same for every $z_\ell \in Z_\ell$, so does $\Pi_{t,r}^{(\ell'')}(z_L, \dots, z_{\ell+1}, z_\ell, \tilde{z}_{\ell-1}, \dots, \tilde{z}_r)$. This completes the induction and finish the proof. $\square$

Now we can use a greedy selection strategy. We upper bound the total number of $\Phi^{(\ell+1)}$ and show that there exists at least one $\tilde{\Phi}^{(\ell+1)}$ such that $S(\tilde{\Phi}^{(\ell+1)})$ has large

size. The proof of the following Lemma can be found at Section IV-D.

Lemma IV.10. *There exists $\tilde{\Phi}^{(\ell+1)}$ such that*

$$|S(\tilde{\Phi}^{(\ell+1)})| \geq |A_L| \cdots |A_{\ell+1}| \cdot 2^{-2\sqrt{K} \cdot (x_0 \cdots x_{\ell-1}) \cdot (n_1 \cdots n_{L-1})}$$

This $\tilde{\Phi}^{(\ell+1)}$ provides us with the fixed transcript in Equation (17).

Given Lemma IV.10, we fix the transcript from players $j \in [\ell+1 : L]$ to players $i \in [-1 : \ell-1]$ at the $(\ell+1)$ -th epoch using $\tilde{\Phi}^{(\ell+1)}$, i.e.,

$$\begin{aligned} \Lambda_{j,i}^{(\ell+1,\ell+1)}(\tilde{z}_\ell, \dots, \tilde{z}_i) &= \tilde{\Phi}_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \\ \text{for } \forall j &\in [\ell+1 : L], i \in [-1 : \ell-1], \\ \tilde{z}_\ell &\in Z_\ell, \dots, \tilde{z}_i \in Z_i. \end{aligned} \quad (17)$$

We set $T_{\geq \ell+1} = S(\tilde{\Phi}^{(\ell+1)}) \subseteq S_{\geq \ell+1}$. We have

Lemma IV.11. *The set $T_{\geq \ell+1}$ is consistent with $(\Lambda_{j,i}^{(\ell+1,\ell')})_{i \in [-1:\ell-1], j \in [\ell+1:L], \ell' \in [\ell+1]}$. Formally, for any $(\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in T_{\geq \ell+1}$ and $\tilde{z}_{<\ell+1} \in Z_{<\ell+1}$, one has*

$$\Pi_{j,i}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_i) = \Lambda_{j,i}^{(\ell+1,\ell')}(\tilde{z}_\ell, \dots, \tilde{z}_i). \quad (18)$$

for any $j \in [\ell+1 : L], i \in [-1 : \ell-1], \ell' \in [\ell+1]$. Moreover, we have

$$|T_{\geq \ell+1}| \geq |A_L| \cdots |A_{\ell+1}| \cdot 2^{-2\sqrt{K} \cdot (x_0 \cdots x_{\ell-1}) \cdot (n_1 \cdots n_{L-1})}. \quad (19)$$

Proof. Since $T_{\geq \ell+1} \subseteq S_{\geq \ell+1}$, by Lemma IV.8, Eq. (18) holds for any $\ell' \in [\ell]$. By Eq. (16) and Eq. (17), we also know that

$$\begin{aligned} \Pi_{j,i}^{(\ell+1)}(\tilde{z}_L, \dots, \tilde{z}_i) &= \tilde{\Phi}_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \\ &= \Lambda_{j,i}^{(\ell+1,\ell+1)}(\tilde{z}_\ell, \dots, \tilde{z}_i) \end{aligned}$$

This completes the proof of Eq. (18). Eq. (19) follows directly from Lemma IV.10. $\square$

4) *Step 2.3: Fixing the transcript to player ℓ :* Finally, we fix the transcript from the player $j \in [\ell+1 : L]$ to the player ℓ at the first $(\ell+1)$ epochs. This follows from the a greedy selection strategy. Let

$$\begin{aligned} \Psi &= \left(\Psi_{j,\ell}^{(\ell')}(\tilde{z}_\ell) \right)_{j \in [\ell+1:L], \ell' \in [\ell+1], \tilde{z}_\ell \in Z_\ell} \\ \text{where } \Psi_{j,\ell}^{(\ell')}(\tilde{z}_\ell) &\in \text{domain}(\Pi_{j,\ell}^{(\ell')}) \end{aligned}$$

Define

$$T(\Psi) := \left\{ \begin{array}{l} (\tilde{z}_L, \dots, \tilde{z}_{\ell+1}) \in T_{\geq \ell+1} : \\ \Pi_{j,\ell}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_\ell) = \Psi_{j,\ell}^{(\ell')}(\tilde{z}_\ell) \quad \forall \tilde{z}_\ell \in Z_\ell, \\ \ell' \in [\ell+1], j \in [\ell+1 : L] \end{array} \right\} \quad (20)$$

We can upper bound the number of different Ψ , and use an average argument to obtain the following Lemma. Its proof can be found at Section IV-D.

Lemma IV.12. *There exists $\tilde{\Psi}$ such that $|T(\tilde{\Psi})| \geq |A_L| \cdots |A_{\ell+1}| / \Delta_{\ell+1}$.*

Given Lemma IV.12, we fix the transcripts from players $j \in [\ell+1 : L]$ to players ℓ at the first $(\ell+1)$ -th epochs using $\tilde{\Psi}$. In particular, we take

$$\begin{aligned} \Lambda_{j,\ell}^{(\ell+1,\ell')}(\tilde{z}_\ell) &= \tilde{\Psi}_{j,\ell}^{(\ell')}(\tilde{z}_\ell) \quad \forall j \in [\ell+1 : L], \\ \ell' &\in [\ell+1], \tilde{z}_i \in Z_i. \end{aligned} \quad (21)$$

We take $R_{\geq \ell+1} = T(\tilde{\Psi})$ and we have

Lemma IV.13. *$R_{\geq \ell+1}$ is consistent with $\Lambda^{(\ell+1)}$. Formally, for any $\tilde{z}_{\geq \ell+1} \in R_{\geq \ell+1}$ and $\tilde{z}_{<\ell+1} \in Z_{<\ell+1}$, one has*

$$\Pi_{j,i}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_i) = \Lambda_{j,i}^{(\ell+1,\ell')}(\tilde{z}_\ell, \dots, \tilde{z}_i). \quad (22)$$

holds for any $j \in [\ell+1 : L], i \in [-1 : \ell], \ell' \in [\ell+1]$. Moreover, we have

$$|R_{\geq \ell+1}| \geq |A_L| \cdots |A_{\ell+1}| / \Delta_{\ell+1} \quad (23)$$

Proof. Since $R_{\geq \ell+1} \subseteq T_{\geq \ell+1}$, by Lemma IV.11, it is consistent with $\{\Lambda_{j,i}^{(\ell+1,\ell')}\}_{j \in [\ell+1:L], i \in [-1:\ell-1], \ell' \in [\ell+1]}$. Combining Eq. (20)(21), we have

$$\Pi_{j,\ell}^{(\ell')}(\tilde{z}_L, \dots, \tilde{z}_\ell) = \Psi_{j,\ell}^{(\ell')}(\tilde{z}_\ell) = \Lambda_{j,\ell}^{(\ell+1,\ell')}(\tilde{z}_\ell).$$

This proves Eq. (22). Eq. (23) follows directly from Lemma IV.12. $\square$

5) *Wrap up the induction:* Combining the above three steps (Eq. (15)(17)(21)), we have fixed the transcript

$$\Lambda^{(\ell+1)} = \{\Lambda_{j,i}^{(\ell+1,\ell')}\}_{j \in [\ell+1:L], i \in [-1:\ell], \ell' \in [\ell+1]}.$$

Now we can wrap up the induction. We need to verify the two inductive hypothesis, i.e., Eq. (9)(10).

For the first inductive hypothesis, by Lemma IV.13, we know that $R_{\geq \ell+1}$ is consistent with $\Lambda^{(\ell+1)}$.

For the second inductive hypothesis (i.e., Eq. (9)), we have proved in Lemma IV.13 that $|R_{\geq \ell+1}| \geq |A_L| \cdots |A_{\ell+1}| / \Delta_{\ell+1}$.

For the last inductive hypothesis (i.e., Eq. (10)), recall we take $Z_\ell = S_2^{(\ell)}$ in Lemma IV.7, we have that

$$\begin{aligned} |\mathcal{I}_\ell| &= |\{i_\ell : i_\ell = \tilde{z}_\ell(\tilde{w}_{\ell-1}, \tilde{i}_{\ell-1}) \\ &\quad \text{for some } \tilde{w}_{\ell-1} \in [n_{\ell-1}], \tilde{i}_{\ell-1} \in \mathcal{I}_{\ell-1}, \tilde{z}_\ell \in S_2^{(\ell)}\}| \\ &\geq \Theta_\ell \end{aligned}$$

This proves Eq. (10)) and completes the induction step.

D. Proof of remaining technical lemmas

Finally, we provide the missing proofs for the technical lemmas used before.

We first prove Lemma IV.6.

Proof of Lemma IV.6. Consider the following greedy approach. Initially, we set $D_0 = \emptyset$. For $\tau = 1, 2, \dots$, we add $\zeta_\tau \in S$ to $D_\tau = D_{\tau-1} \cup \{\zeta_\tau\}$ if it satisfies

$$|\{\zeta_\tau(i_0) : i_0 \in Z_0\} \setminus \{\tilde{z}_1(i_0) : \tilde{z}_1 \in D_{\tau-1}, i_0 \in Z_0\}| \geq 8^{-L}x_0. \quad (24)$$

If the process continues to $\tau = x_1 + 1$, then we take $Z_1 = D_{x_1}$, we have $|Z_1| = |D_{x_1}| = x_1$ and by Eq. (24), it satisfies the requirement of Eq. (12). If the process stops for $\tau \leq x_1$ and $|\{\tilde{z}_1(i_0) : \tilde{z}_1 \in D_{\tau-1}, i_0 \in Z_0\}| \geq 8^{-L}x_0x_1$, then we can take Z_1 to be the union of $D_{\tau-1}$ and arbitrary $x_1 - \tau + 1$ elements from S , we have $|Z_1| = x_1$ and it satisfies the requirement of Eq. (12).

Otherwise, suppose it stops for $\tau \leq x_1$ and $|\{\tilde{z}_1(i_0) : \tilde{z}_1 \in D_{\tau-1}, i_0 \in Z_0\}| < 8^{-L}x_0x_1$. We claim that for every $\xi \in S$, its entries $\xi(1), \dots, \xi(x_0)$ take value from either $\{\tilde{z}_1(i_0) : \tilde{z}_1 \in D_{\tau-1}, i_0 \in Z_0\}$, or a subset from $[N_0] \setminus \{\tilde{z}_1(i_0) : \tilde{z}_1 \in D_{\tau-1}, i_0 \in Z_0\}$ of size at most $8^{-L}x_0$, the total number of such ζ is at most

$$\begin{aligned} & \binom{N_0}{8^{-L}x_0} \cdot (8^{-L}x_0x_1)^{x_0} \cdot (N_0)^{N_0-x_0} \\ & \leq |A_0| \cdot (m^{8^{-L}} \cdot 8^{-L}x_0x_1/m)^{x_0} \\ & < |A_0| \cdot 2^{-2Hdp \cdot (n_1 \dots n_{L-1})} \end{aligned}$$

The first step follows from $A_0 = N_0^{N_0}$, $N_0 = m$, the second step follows from (1) $m^{8^{-L}} \cdot 8^{-L}x_0x_1 \leq m/2$ and $x_0 > 2Hdp(n_1 \dots n_{L-1})$ (see Eq. (4)(7) for the choice of parameters). This contradicts with Eq. (11) and completes the proof. $\square$

The proof of Lemma IV.7 involves several intermediate steps, and we present it here.

Proof of Lemma IV.7. We simplify the notation a bit in the proof. We write

$$\begin{aligned} R &:= R_{\geq \ell}, \quad \mathcal{A} := A_L \times \dots \times A_{\ell+1}, \quad \mathcal{B} := A_\ell, \\ \Delta &:= \Delta_\ell, \quad x := x_\ell. \end{aligned} \quad (25)$$

For any $b \in \mathcal{B}$, let

$$R_b = \{a \in \mathcal{A} : (a, b) \in R\} \subseteq \mathcal{A}$$

By Eq. (9)(25), we have

$$\sum_{b \in \mathcal{B}} |R_b| = |R| \geq |\mathcal{A}| |\mathcal{B}| / \Delta \quad (26)$$

For any $b \in \mathcal{B} = [N_{\ell-1}]^{N_{\ell-1}}$, define

$$\begin{aligned} I(b) &:= \{b(\tilde{w}_{\ell-1}, \tilde{i}_{\ell-1}) : \tilde{w}_{\ell-1} \in [n_{\ell-1}], \tilde{i}_{\ell-1} \in \mathcal{I}_{\ell-1}\} \\ &\subseteq [N_{\ell-1}] \end{aligned}$$

The key step is to prove

Lemma IV.14. *There exists two sequences $b_1, \dots, b_x \in \mathcal{B}$ and $\mathcal{A}_1, \dots, \mathcal{A}_x \subseteq \mathcal{A}$ that satisfy*

- $\mathcal{A}_x \subseteq \mathcal{A}_{x-1} \dots \subseteq \mathcal{A}_1 \subseteq \mathcal{A}_0 = \mathcal{A}$, and $|\mathcal{A}_t| \geq |\mathcal{A}_{t-1}|/\Delta^2$ ($t \in [x]$)
- $\mathcal{A}_t \subseteq R_{b_t}$ ($t \in [x]$)
- $|I(b_t) \setminus \cup_{t' \in [t-1]} I(b_{t'})| \geq 8^{-L}n_{\ell-1}|\mathcal{I}_{\ell-1}|$ ($t \in [x]$)
- $\sum_{b \in \mathcal{B}} |R_b \cap \mathcal{A}_t| \geq (1 - \frac{1}{x})^t \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta}$ ($t \in [0 : x]$)

With Lemma IV.14, we can easily finish the proof of Lemma IV.7. To see this, we take

$$S_1^{(\ell)} = \mathcal{A}_x \quad \text{and} \quad S_2^{(\ell)} = \{b_1, \dots, b_x\}.$$

Then we have $|S_1^{(\ell)}| \geq |\mathcal{A}|/\Delta^{2x}$ due to the first claim in Lemma IV.14 and $|S_2^{(\ell)}| = x$. Moreover, by the second claim in Lemma IV.14, we have $S_1^{(\ell)} = \mathcal{A}_x \subseteq R_b$ for any $b \in S_2^{(\ell)}$, hence we have $S^\ell = S_1^{(\ell)} \times S_2^{(\ell)} \subseteq R$.

For the second property of Lemma IV.7, we have

$$\begin{aligned} & |\{i_\ell : i_\ell = \tilde{z}_\ell(\tilde{w}_{\ell-1}, \tilde{i}_{\ell-1}) \\ & \quad \text{for some } \tilde{w}_{\ell-1} \in [n_{\ell-1}], \tilde{i}_{\ell-1} \in \mathcal{I}_{\ell-1}, \tilde{z}_\ell \in S_2^{(\ell)}\}| \\ & = |\cup_{t \in [x]} I(b_x)| \geq 8^{-L}x_\ell n_{\ell-1} |\mathcal{I}_{\ell-1}| \\ & \geq 8^{-L}x_\ell n_{\ell-1} \Theta_{\ell-1} \\ & \geq \Theta_\ell \end{aligned}$$

The second step follows from the third claim of Lemma IV.14, the third step follows from the inductive hypothesis of Lemma IV.5 (Eq. (10)), the last step follows from the definition of Θ_ℓ (see Eq. (8)). This completes the proof of Lemma IV.7. $\square$

We next provide the proof of the key Lemma.

Proof of Lemma IV.14. We prove by induction. The case of $t = 0$ follows directly from Eq. (26). Suppose the claim holds up to t , then for $t + 1$, let

$$R'_b := R_b \cap \mathcal{A}_t \quad \forall b \in \mathcal{B}.$$

By the fourth induction hypothesis, we have

$$\sum_{b \in \mathcal{B}} |R'_b| = \sum_{b \in \mathcal{B}} |R_b \cap \mathcal{A}_t| \geq (1 - \frac{1}{x})^t \cdot \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta}. \quad (27)$$

Consider the set

$$J_t := \cup_{t' \in [t]} I(b_{t'})$$

its size satisfies

$$8^{-L}tn_{\ell-1}|\mathcal{I}_{\ell-1}| \leq |J_t| \leq tn_{\ell-1}|\mathcal{I}_{\ell-1}|$$

where the LHS follows from the third inductive hypothesis, i.e., $|J_t| = \sum_{t' \in [t]} |J_{t'} \setminus J_{t'-1}| \geq t \cdot 8^{-L}n_{\ell-1}|\mathcal{I}_{\ell-1}|$, the RHS holds since $|J_t| \leq \sum_{t' \in [t]} |I(b_{t'})| \leq t \cdot n_{\ell-1}|\mathcal{I}_{\ell-1}|$.

Define

$$C := \{b \in \mathcal{B} : |I(b) \setminus J_t| < 8^{-L} n_{\ell-1} |\mathcal{I}_{\ell-1}|\} \subseteq \mathcal{B} \quad (28)$$

We have the following claim, whose proof is deferred till the end.

Lemma IV.15. *We have $|C| \leq \frac{|\mathcal{B}|}{\Delta^2}$.*

Define

$$R''_b := \begin{cases} R'_b & b \notin C \\ \emptyset & b \in C \end{cases} \quad \forall b \in \mathcal{B}. \quad (29)$$

That is, R''_b is set to empty if $I(b)$ does not differ too much from J_t .

By Lemma IV.15, we have

$$\begin{aligned} \sum_{b \in \mathcal{B}} |R''_b| &= \sum_{b \in \mathcal{B}} |R'_b| - \sum_{b \in C} |R'_b| \\ &\geq \left(1 - \frac{1}{x}\right)^t \cdot \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta} - \frac{|\mathcal{B}|}{\Delta^2} \cdot |\mathcal{A}_t|. \end{aligned} \quad (30)$$

The first step follows from the definition of R''_b (see Eq. (29)), the second step follows from Eq. (27), Lemma IV.15 and $|R'_b| \leq |\mathcal{A}_t|$.

Now we are going to select $b_{t+1} \in \mathcal{B}$ and $\mathcal{A}_{t+1} \subseteq \mathcal{A}_t$. To this end, consider the following greedy process. Initially, set $D = \emptyset$ and $\tau = 0$. If there exists $\beta_\tau \in \mathcal{B}$ such that

$$R''_{\beta_\tau} \setminus \left(\bigcup_{\beta \in D} R''_\beta \right) \geq \frac{|\mathcal{A}_t|}{\Delta^2}, \quad (31)$$

then define

$$O_\tau := R''_{\beta_\tau} \setminus \left(\bigcup_{\beta \in D} R''_\beta \right) \subseteq \mathcal{A}_t \quad (32)$$

and update $D \leftarrow D \cup \{\beta_\tau\}$ and $\tau = \tau + 1$. It is clearly that D is not empty (see Eq (30)) and the process would stop. Let

$$O := \bigcup_{\tau \in [0:|D|-1]} O_\tau \quad \text{and} \quad R'''_b := R''_b \cap O \quad \forall b \in \mathcal{B},$$

we have

$$\begin{aligned} \sum_{b \in \mathcal{B}} |R'''_b| &= \sum_{b \in \mathcal{B}} |R''_b \cap O| \\ &= \sum_{b \in \mathcal{B}} |R''_b| - \sum_{b \in \mathcal{B}} |R''_b \setminus O| \\ &\geq \left(1 - \frac{1}{x}\right)^t \cdot \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta} - \frac{|\mathcal{B}|}{\Delta^2} \cdot |\mathcal{A}_t| - |\mathcal{B}| \cdot \frac{|\mathcal{A}_t|}{\Delta^2} \\ &\geq \left(1 - \frac{1}{x}\right)^{t+1} \cdot \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta} \end{aligned}$$

Here the third step follows from Eq. (30) and the selection strategy (i.e. Eq. (31)), the last step holds since $\Delta \gg x$.

Since $\{O_\tau\}_{\tau \in [0:|D|-1]}$ are disjoint, we claim that, there exists $\tau \in [0:|D|-1]$, such that

$$\begin{aligned} \sum_{b \in \mathcal{B}} |R'''_b \cap O_\tau| &\geq \left(1 - \frac{1}{x}\right)^{t+1} \cdot \frac{|\mathcal{A}_t| |\mathcal{B}|}{\Delta} \cdot \frac{|O_\tau|}{|O|} \\ &\geq \left(1 - \frac{1}{x}\right)^{t+1} \frac{|O_\tau| |\mathcal{B}|}{\Delta}. \end{aligned} \quad (33)$$

Here the second step follows from $|\mathcal{A}_t| \geq |O|$.

We take $\mathcal{A}_{t+1} = O_\tau$, $b_{t+1} = \beta_\tau$, and we prove the four inductive hypothesis continue to holds.

First, by Eq. (31)(32), we have $\mathcal{A}_{t+1} = O_\tau \subseteq \mathcal{A}_t$, and

$$|\mathcal{A}_{t+1}| = |O_\tau| \geq \frac{|\mathcal{A}_t|}{\Delta^2} \quad (34)$$

Second, by Eq. (32), we have

$$\mathcal{A}_{t+1} = O_\tau \subseteq R''_{\beta_\tau} = R''_{b_{t+1}} \subseteq R_{b_{t+1}} \quad (35)$$

Third, it is clear that $R''_{b_{t+1}}$ is not empty, this implies that $b_{t+1} \notin C$, and therefore

$$|I(b_{t+1}) \setminus J_t| \geq 8^{-L} \cdot n_{\ell-1} |\mathcal{I}_{\ell-1}|. \quad (36)$$

Finally, by Eq (33), we have

$$\begin{aligned} \sum_{b \in \mathcal{B}} |R_b \cap \mathcal{A}_{t+1}| &= \sum_{b \in \mathcal{B}} |R_b \cap O_\tau| \\ &\geq \sum_{b \in \mathcal{B}} |R'_b \cap O_\tau| \\ &\geq \left(1 - \frac{1}{x}\right)^{t+1} \frac{|O_\tau| |\mathcal{B}|}{\Delta} \\ &= \left(1 - \frac{1}{x}\right)^{t+1} \frac{|\mathcal{A}_{t+1}| |\mathcal{B}|}{\Delta}. \end{aligned} \quad (37)$$

Combining Eq. (34)(35)(36)(37), we complete the proof. $\square$

Proof of Lemma IV.15. We compute the size of C . For any $b \in C \subseteq \mathcal{A}_\ell = [N_{\ell-1}]^{N_{\ell-1}}$, one can view it as a mapping from $[N_{\ell-1}]$ to $[N_{\ell-1}]$. By Eq. (28), $b \in C$ if and only if it does not differ much than J_t on entries $(w_{\ell-1}, i_{\ell-1}) \in [n_{\ell-1}] \times \mathcal{I}_{\ell-1}$. For entries in $[N_\ell] \setminus ([n_{\ell-1}] \times \mathcal{I}_{\ell-1})$, it poses no constraints so it has $(N_{\ell-1})^{N_{\ell-1} - n_{\ell-1} |\mathcal{I}_{\ell-1}|}$ number of choices.

For entries in $[n_{\ell-1}] \times \mathcal{I}_{\ell-1}$, it could take value from J_t , or from a subset of at size at most $8^{-L} n_{\ell-1} |\mathcal{I}_{\ell-1}|$ in $[N_\ell] \setminus ([n_{\ell-1}] \times \mathcal{I}_{\ell-1})$, the total number of choices is at most

$$\begin{aligned} &\binom{N_{\ell-1}}{8^{-L} n_{\ell-1} |\mathcal{I}_{\ell-1}|} \cdot (x n_{\ell-1} |\mathcal{I}_{\ell-1}|)^{n_{\ell-1} |\mathcal{I}_{\ell-1}|} \\ &\leq (N_{\ell-1}^{8^{-L}} \cdot x n_{\ell-1} |\mathcal{I}_{\ell-1}|)^{n_{\ell-1} |\mathcal{I}_{\ell-1}|} \end{aligned}$$

In summary, the size of C is at most

$$\begin{aligned} & (N_{\ell-1})^{N_{\ell-1}-n_{\ell-1}|\mathcal{I}_{\ell-1}|} \cdot (N_{\ell-1}^{8-L} \cdot x n_{\ell-1} |\mathcal{I}_{\ell-1}|)^{n_{\ell-1}|\mathcal{I}_{\ell-1}|} \\ &= |\mathcal{B}| \cdot (N_{\ell-1}^{8-L} \cdot x n_{\ell-1} |\mathcal{I}_{\ell-1}| / N_{\ell-1})^{n_{\ell-1}|\mathcal{I}_{\ell-1}|} \\ &\leq |\mathcal{B}| \cdot (1/2)^{n_{\ell-1}|\mathcal{I}_{\ell-1}|} \\ &\leq |\mathcal{B}| / \Delta^2 \end{aligned}$$

The first step follows from the definition of x and $\mathcal{B}$. For the second step, it follows from

$$\begin{aligned} N_{\ell-1}^{8-L} \cdot x n_{\ell-1} |\mathcal{I}_{\ell-1}| &\leq \sqrt{K} \cdot x n_{\ell-1} (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{\ell-2}) \\ &\leq \frac{1}{2} m \cdot (n_1 \dots n_{\ell-1}) = \frac{1}{2} N_{\ell-1} \end{aligned}$$

where we use the fact that $N_{\ell-1}^{8-L} \leq \sqrt{K}$, $|\mathcal{I}_{\ell-1}| \leq (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{\ell-2})$ in the first step, the second step follows from the choice of parameters (see Eq. (4)(7)).

For the third step, we have

$$\begin{aligned} \Delta^2 &= \Delta_{\ell}^2 = 2^{8\sqrt{K}(x_0 \dots x_{\ell-2}) \cdot (n_1 \dots n_{L-1})} \\ &\leq 2^{8^{-L\ell} \cdot n_{\ell-1} (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{\ell-2})} \leq 2^{n_{\ell-1} |\mathcal{I}_{\ell-1}|} \end{aligned}$$

The second step follows from the definition of Δ_{ℓ} (see Eq. (8)), the third step follows from the choice of parameters Eq. (4)(7). The last step follows from the induction hypothesis of Lemma IV.5. $\square$

We next prove Lemma IV.10. First, we need

Lemma IV.16. *The total number of possible $\Phi^{(\ell+1)}$ is at most $2^{\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}$.*

Proof. Recall

$$\Phi^{(\ell+1)} = \left(\Phi_{j,i}^{(\ell+1)} \right)_{j \in [\ell+1:L], i \in \{e\} \cup [0:\ell-1]}$$

and

$$\begin{aligned} \Phi_{j,i}^{(\ell+1)} &= \left(\Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) \right)_{\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i} \\ \text{and } \Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i) &\in \text{domain}(\Pi_{j,i}^{(\ell+1)}). \end{aligned}$$

For $i \in [\ell-1]$, $\Phi_{j,i}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_i)$ takes value from $\{0, 1\}^{2Hdp \cdot N_{i-1}}$, summing over $\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_i \in Z_i$, $j \in [\ell+1:L]$, the total number of $(\Phi_{j,i}^{(\ell+1)})_{j \in [\ell+1:L]}$ is at most

$$(2^{2Hdp \cdot N_{i-1}})^{x_i \dots x_{\ell-1} \cdot L} \leq 2^{2HdpL \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}$$

where the inequality follows from the choice of parameters (see Eq. (4)(5)(7)).

For $i = 0$, $\Phi_{j,0}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_0)$ takes value from $\{0, 1\}^{2Hdp}$, summing over $\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_0 \in Z_0$ and $j \in [\ell+1:L]$, the total number of $(\Phi_{j,0}^{(\ell+1)})_{j \in [\ell+1:L]}$ is at most

$$(2^{2Hdp})^{x_0 \dots x_{\ell-1} \cdot L} \leq 2^{2HdpL \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}.$$

For $i = -1$, $\Phi_{j,-1}^{(\ell+1)}(\tilde{z}_{\ell-1}, \dots, \tilde{z}_0, \tilde{z}_{-1})$ takes value from $\{0, 1\}^{2Hdp}$, summing over $\tilde{z}_{\ell-1} \in Z_{\ell-1}, \dots, \tilde{z}_0 \in Z_0, \tilde{z}_{-1} \in [Z_{-1}]$ and $j \in [\ell+1:L]$, $(\Phi_{j,-1}^{(\ell+1)})_{j \in [\ell+1:L]}$ is at most

$$(2^{2Hdp})^{(x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1}) \cdot L} = 2^{2HdpL \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}$$

Taking a product over all these terms, the total number of $\Phi^{(\ell+1)}$ is bounded by

$$\begin{aligned} & \left(2^{2HdpL \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \right)^L \\ & \leq 2^{\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \end{aligned}$$

$\square$

Now we can finish the proof of Lemma IV.10.

Proof of Lemma IV.10. By Lemma IV.9 and Lemma IV.7, we have that

$$\sum_{\Phi^{(\ell+1)}} |S(\Phi^{(\ell+1)})| = |S_{\geq \ell+1}| \geq \frac{|A_L| \dots |A_{\ell+1}|}{\Delta_{\ell}^{2x_{\ell}}}.$$

By Lemma IV.16, there exists $\tilde{\Phi}^{(\ell+1)}$, such that

$$\begin{aligned} |S(\tilde{\Phi}^{(\ell+1)})| &\geq \frac{|A_L| \dots |A_{\ell+1}|}{\Delta_{\ell}^{2x_{\ell}}} \cdot 2^{-\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \\ &\geq |A_L| \dots |A_{\ell+1}| \cdot 2^{-2\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \end{aligned}$$

Here the second follows from the choice of parameters (see Eq. (4)(7)(8)). $\square$

We next prove Lemma IV.12

Proof of Lemma IV.12. First, we have $\cup_{\Psi} T(\Psi) = T_{\geq \ell+1}$ and by Lemma IV.11, we have

$$\begin{aligned} \sum_{\Psi} |T(\Psi)| &= |T_{\geq \ell+1}| \\ &\geq |A_L| \dots |A_{\ell+1}| \cdot 2^{-2\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}. \end{aligned}$$

Moreover, the total number of Ψ is at most

$$(2^{2Hdp \cdot N_{\ell-1}})^{x_{\ell} \cdot L^2} \leq 2^{\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})}.$$

Here the equality holds due to the choice of parameters (see Eq. (4)(5)(7)).

Hence, there exists $\tilde{\Psi}$ such that

$$\begin{aligned} & |T(\tilde{\Psi})| \\ & \geq |A_L| \dots |A_{\ell+1}| \cdot 2^{-2\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \\ & \quad \cdot 2^{-\sqrt{K} \cdot (x_0 \dots x_{\ell-1}) \cdot (n_1 \dots n_{L-1})} \\ & \geq |A_L| \dots |A_{\ell+1}| / \Delta_{\ell+1} \end{aligned}$$

The last step follows from the definition of $\Delta_{\ell+1}$ (see Eq. (8)). This completes the proof. $\square$

REFERENCES

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altmerschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 1, learning hierarchical language structures. *ArXiv e-prints, abs/2305.13673*, May, 2023.
- [3] Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 3.1, knowledge storage and extraction. *arXiv preprint arXiv:2309.14316*, 2023.
- [4] Eric Allender and Michal Koucký. Amplifying lower bounds by means of self-reducibility. *J. ACM*, 57(3):14:1–14:36, 2010.
- [5] Josh Alman, Timothy M. Chan, and R. Ryan Williams. Polynomial representations of threshold functions and algorithmic applications. In Irit Dinur, editor, *IEEE 57th Annual Symposium on Foundations of Computer Science, FOCS 2016, 9-11 October 2016, Hyatt Regency, New Brunswick, New Jersey, USA*, pages 467–476. IEEE Computer Society, 2016.
- [6] Alireza Amiri, Xinting Huang, Mark Rofin, and Michael Hahn. Lower bounds for chain-of-thought reasoning in hard-attention transformers. *arXiv preprint arXiv:2502.02393*, 2025.
- [7] Sanjeev Arora and Anirudh Goyal. A theory for emergence of complex skills in language models. *arXiv preprint arXiv:2307.15936*, 2023.
- [8] Sepehr Assadi and Vishvajeet N. Graph streaming lower bounds for parameter estimation and property testing via a streaming XOR lemma. In Samir Khuller and Virginia Vassilevska Williams, editors, *STOC '21: 53rd Annual ACM SIGACT Symposium on Theory of Computing, Virtual Event, Italy, June 21-25, 2021*, pages 612–625. ACM, 2021.
- [9] Guy Bresler and Dheeraj Nagaraj. Sharp representation theorems for relu networks with precise dependence on depth. *Advances in Neural Information Processing Systems*, 33:10697–10706, 2020.
- [10] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, et al. Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712*, 2023.
- [11] Lijie Chen and Roei Tell. Bootstrapping results for threshold circuits “just beyond” known lower bounds. In Moses Charikar and Edith Cohen, editors, *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing, STOC 2019, Phoenix, AZ, USA, June 23-26, 2019*, pages 34–41. ACM, 2019.
- [12] Ruiwen Chen, Rahul Santhanam, and Srikanth Srinivasan. Average-case lower bounds and satisfiability algorithms for small threshold circuits. *Theory Comput.*, 14(1):1–55, 2018.
- [13] David Chiang and Peter Cholak. Overcoming a theoretical limitation of self-attention. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7654–7664, 2022.
- [14] Amit Daniely. Depth separation for neural networks. In *Conference on Learning Theory*, pages 690–696. PMLR, 2017.
- [15] Nouha Dziri, Ximing Lu, Melanie Sclar, Xiang Lorraine Li, Liwei Jiang, Bill Yuchen Lin, Peter West, Chandra Bhagavatula, Ronan Le Bras, Jena D Hwang, et al. Faith and fate: Limits of transformers on compositionality. *Advances in neural information processing systems*, 2023.
- [16] Javid Ebrahimi, Dhruv Gelda, and Wei Zhang. How can self-attention networks recognize dyck-n languages? In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4301–4306, 2020.
- [17] Ronen Eldan and Ohad Shamir. The power of depth for feedforward neural networks. In *Conference on learning theory*, pages 907–940. PMLR, 2016.
- [18] Ethan Ewer, Daewon Chae, Thomas Zeng, Jinkyu Kim, and Kangwook Lee. Entp: Encoder-only next token prediction. *arXiv preprint arXiv:2410.01600*, 2024.
- [19] Guhao Feng, Yuntian Gu, Bohang Zhang, Haotian Ye, Di He, and Liwei Wang. Towards revealing the mystery behind chain of thought: a theoretical perspective. *Advances in neural information processing systems*, 2023.
- [20] Zihao Fu, Wai Lam, Qian Yu, Anthony Man-Cho So, Shengding Hu, Zhiyuan Liu, and Nigel Collier. Decoder-only or encoder-decoder? interpreting language model as a regularized encoder-decoder. *arXiv preprint arXiv:2304.04052*, 2023.
- [21] Michael Hahn. Theoretical limitations of self-attention in neural sequence models. *Transactions of the Association for Computational Linguistics*, 8:156–171, 2020.
- [22] Pooya Hatami, William M. Hoza, Avishay Tal, and Roei Tell. Depth-d threshold circuits vs. depth-(d+1) AND-OR trees. In Barna Saha and Rocco A. Servedio, editors, *Proceedings of the 55th Annual ACM Symposium on Theory of Computing, STOC 2023, Orlando, FL, USA, June 20-23, 2023*, pages 895–904. ACM, 2023.
- [23] John Hewitt, Michael Hahn, Surya Ganguli, Percy Liang, and Christopher D Manning. Rnns can generate bounded hierarchical languages with optimal memory. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1978–2010, 2020.
- [24] Russell Impagliazzo, Ramamohan Paturi, and Michael E. Saks. Size-depth tradeoffs for threshold circuits. *SIAM J. Comput.*, 26(3):693–707, 1997.
- [25] Samy Jelassi, David Brandfonbrener, Sham M Kakade, et al. Repeat after me: Transformers are better than state space models at copying. In *Forty-first International Conference on Machine Learning*, 2024.
- [26] Samy Jelassi, Clara Mohri, David Brandfonbrener, Alex Gu, Nikhil Vyas, Nikhil Anand, David Alvarez-Melis, Yuanzhi Li, Sham M Kakade, and Eran Malach. Mixture of parrots: Experts improve memorization more than reasoning. *arXiv preprint arXiv:2410.19034*, 2024.
- [27] Diederik P Kingma and Jimmy Ba. Adam: A method for stochastic optimization. *arXiv preprint arXiv:1412.6980*, 2014.
- [28] Yoav Levine, Noam Wies, Or Sharir, Hofit Bata, and Amnon Shashua. Limits to depth efficiencies of self-attention. *Advances in Neural Information Processing Systems*, 33:22640–22651, 2020.
- [29] Zhiyuan Li, Hong Liu, Denny Zhou, and Tengyu Ma. Chain of thought empowers transformers to solve inherently serial problems. In *International Conference on Learning Representations*, 2024.
- [30] Shiyu Liang and Rayadurgam Srikant. Why deep neural networks for function approximation? In *International Conference on Learning Representations*, 2017.
- [31] Bingbin Liu, Jordan T Ash, Surbhi Goel, Akshay Krishnamurthy, and Cyril Zhang. Transformers learn shortcuts to automata. In *The Eleventh International Conference on Learning Representations*, 2024.
- [32] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.
- [33] Zhou Lu, Hongming Pu, Feicheng Wang, Zhiqiang Hu, and Liwei Wang. The expressive power of neural networks: A view from the width. *Advances in neural information processing systems*, 30, 2017.
- [34] Ben Mann, N Ryder, M Subbiah, J Kaplan, P Dhariwal, A Neelakantan, P Shyam, G Sastry, A Askell, S Agarwal, et al. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 1, 2020.
- [35] James Martens, Arkadev Chattopadhyay, Toni Pitassi, and Richard Zemel. On the representational efficiency of restricted boltzmann machines. *Advances in Neural Information Processing Systems*, 26, 2013.
- [36] William Merrill and Ashish Sabharwal. The expressive power of transformers with chain of thought. *arXiv preprint arXiv:2310.07923*, 2023.
- [37] William Merrill and Ashish Sabharwal. The parallelism tradeoff: Limitations of log-precision transformers. *Transactions of the Association for Computational Linguistics*, 11:531–545, 2023.
- [38] Dan Saattrup Nielsen, Kenneth Enevoldsen, and Peter Schneider-Kamp. Encoder vs decoder: Comparative analysis of encoder

- and decoder language models on multilingual nlu tasks. *arXiv preprint arXiv:2406.13469*, 2024.
- [39] Noam Nisan and Avi Wigderson. Rounds in communication complexity revisited. *SIAM J. Comput.*, 22(1):211–219, 1993.
- [40] Binghui Peng, Srinu Narayanan, and Christos Papadimitriou. On limitations of the transformer architecture. *Conference on Language Modeling*, 2024.
- [41] Jorge Pérez, Pablo Barceló, and Javier Marinkovic. Attention is turing-complete. *Journal of Machine Learning Research*, 22(75):1–35, 2021.
- [42] Jackson Petty, Sjoerd Steenkiste, Ishita Dasgupta, Fei Sha, Dan Garrette, and Tal Linzen. The impact of depth on compositional generalization in transformer language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 7232–7245, 2024.
- [43] Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A Smith, and Mike Lewis. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711, 2023.
- [44] Muhammad Qorib, Geonsik Moon, and Hwee Tou Ng. Are decoder-only language models better than encoder-only language models in understanding word meaning? In *Findings of the Association for Computational Linguistics ACL 2024*, pages 16339–16347, 2024.
- [45] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, Ilya Sutskever, et al. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019.
- [46] Vwani P. Roychowdhury, Alon Orlitsky, and Kai-Yeung Siu. Lower bounds on threshold and related circuits via communication complexity. *IEEE Trans. Inf. Theory*, 40(2):467–474, 1994.
- [47] Itay Safran, Ronen Eldan, and Ohad Shamir. Depth separations in neural networks: what is actually being separated? In *Conference on Learning Theory*, pages 2664–2666. PMLR, 2019.
- [48] Itay Safran and Ohad Shamir. Depth-width tradeoffs in approximating natural functions with neural networks. In *International conference on machine learning*, pages 2979–2987. PMLR, 2017.
- [49] Clayton Sanford, Daniel Hsu, and Matus Telgarsky. Representational strengths and limitations of transformers. *Advances in neural information processing systems*, 2023.
- [50] Clayton Sanford, Daniel Hsu, and Matus Telgarsky. One-layer transformers fail to solve the induction heads task. *arXiv preprint arXiv:2408.14332*, 2024.
- [51] Clayton Sanford, Daniel Hsu, and Matus Telgarsky. Transformers, parallel computation, and logarithmic depth. In *International Conference on Machine Learning*, 2024.
- [52] Rahul Santhanam and Ryan Williams. On medium-uniformity and circuit lower bounds. In *2013 IEEE Conference on Computational Complexity*, pages 15–23. IEEE, 2013.
- [53] Yi Tay. What happened to bert & t5? on transformer encoders, prefixlm and denoising objectives. 2024. <https://www.yitay.net/blog/model-architecture-blogpost-encoders-prefixlm-denoising>.
- [54] Matus Telgarsky. Benefits of depth in neural networks. In *Conference on learning theory*, pages 1517–1539. PMLR, 2016.
- [55] Roei Tell. On implications of better sub-exponential lower bounds for AC0, 2020. Available at: <https://sites.google.com/site/roetell/Expositions>.
- [56] Gal Vardi, Daniel Reichman, Toniann Pitassi, and Ohad Shamir. Size and depth separation in approximating benign functions with neural networks. In *Conference on Learning Theory*, pages 4195–4223. PMLR, 2021.
- [57] Gal Vardi and Ohad Shamir. Neural networks with small weights and depth-separation barriers. *Advances in neural information processing systems*, 33:19433–19442, 2020.
- [58] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.
- [59] Siwei Wang, Yifei Shen, Shi Feng, Haoran Sun, Shang-Hua Teng, and Wei Chen. Alpine: Unveiling the planning capability of autoregressive learning in language models. *Advances in neural information processing systems*, 2024.
- [60] Jason Wei, Yi Tay, Rishi Bommasani, Colin Raffel, Barret Zoph, Sebastian Borgeaud, Dani Yogatama, Maarten Bosma, Denny Zhou, Donald Metzler, et al. Emergent abilities of large language models. *arXiv preprint arXiv:2206.07682*, 2022.
- [61] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Fei Xia, Ed Chi, Quoc V Le, Denny Zhou, et al. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.
- [62] Kaiyue Wen, Xingyu Dang, and Kaifeng Lyu. Rnns are not transformers (yet): The key bottleneck on in-context retrieval. *arXiv preprint arXiv:2402.18510*, 2024.
- [63] Kaiyue Wen, Yuchen Li, Bingbin Liu, and Andrej Risteski. Transformers are uninterpretable with myopic methods: a case study with bounded dyck grammars. *Advances in Neural Information Processing Systems*, 36, 2023.
- [64] Sohee Yang, Elena Gribovskaya, Nora Kassner, Mor Geva, and Sebastian Riedel. Do large language models latently perform multi-hop reasoning? In *Association for Computational Linguistics: ACL-IJCNLP 2024*, 2024.
- [65] Shunyu Yao, Binghui Peng, Christos Papadimitriou, and Karthik Narasimhan. Self-attention networks can process bounded hierarchical languages. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 3770–3785, 2021.
- [66] Dmitry Yarotsky. Error bounds for approximations with deep relu networks. *Neural networks*, 94:103–114, 2017.
- [67] Tian Ye, Zicheng Xu, Yuanzhi Li, and Zeyuan Allen-Zhu. Physics of language models: Part 2.1, grade-school math and the hidden reasoning process. *arXiv preprint arXiv:2407.20311*, 2024.

APPENDIX A

MISSING PROOF FROM SECTION I-B

We sketch the high level proof idea of Corollary I.2, Corollary I.3 and Corollary I.4. Note that the lower bound parts of these corollaries all follow directly from Theorem I.1, so in the following it suffices to prove the upper bound parts.

a) *Retrieval head.*: First, we observe that an attention head could implement the following retrieval operation. Let $i \in [n]$, suppose $x_j^{(\ell)}$ (i.e., the input to the ℓ -th layer at position j) contains $a_j \in \{0, 1\}^D$ and $b_j \in \{0, 1\}^D$ for any previous position $j \leq i - 1$ and suppose $x_i^{(\ell)}$ contains a “query” a . The retrieval task at position i (of layer ℓ) is to find the position j such that $a_j = a$ and return the value of b_j (if there are multiple or no such positions, then the return value could be arbitrary).

This retrieval operation can be implemented with one attention head. In particular, we set the V -value to be b_j , K -value to be $\log^2(n) \cdot (a_j, \vec{1} - a_j)$ for position $j < i$,¹³ the Q -value at position i is taken to be $\log^2(n) \cdot (a, \vec{1} - a)$. The attention score (before softmax) satisfies

$$\langle Qx_i^{(\ell)}, Kx_j^{(\ell)} \rangle = \begin{cases} \log^2(n)D & a_j = a \\ \leq \log^2(n)D - \log^2(n) & a_j \neq a. \end{cases}$$

¹³Here, $\vec{1} \in \{0, 1\}^D$ denotes the all-1 vector of length D , and $\vec{1} - a_j$ denotes element-wise subtraction.

Hence, if there is exactly one position $j \leq i$ that satisfies $a_j = a$, then the attention head would only attend to position j and get the value b_j , assuming the precision $p = \Theta(\log(n))$.

b) *Proof of Corollary 1.2.*: Consider an $(L + 1)$ -layer Transformer such that each layer has one attention head. For any $\ell \in [L + 1]$, let the attention head at layer ℓ implements the retrieval task for the $(\ell - 1)$ -th composition, i.e., given $i_{\ell-2}$ and $w_{\ell-2}$ at the last token, find $i_{\ell-1} = z_{\ell-1}(w_{\ell-2}, i_{\ell-2})$. Concretely, the last token implements the query $a = (w_{\ell-2}, i_{\ell-2}) \in [N_{\ell-2}]$. For every previous token $j \in [n - 1]$, if the j -th token corresponds to the t -th entry of $z_{\ell-1}$ ($t \in [N_{\ell-1}]$), then it sets $a_j = t$ and $b_j = z_{\ell-1}(t)$; otherwise, if the j -th token does not corresponds to any entry of $z_{\ell-1}$, it provides an arbitrary dummy pair of a_j and b_j . One can inductively prove that the last token successfully retrieves the value of $i_{\ell-1}$ after layer ℓ .

c) *The proof of Corollary 1.4.*: The proof is similar to Corollary 1.2. We design $L + 1$ attention heads, where the ℓ -th attention head aims to retrieve $i_{\ell-1} = z_{\ell-1}(w_{\ell-2}, i_{\ell-2})$ given $w_{\ell-2}, i_{\ell-2}$. One can inductively prove that at the ℓ -th CoT step, the Transformer could obtain $i_{\ell-1}$ and include it into the next token.

d) *Proof of Corollary 1.3.*: Recall we want to construct an $O(\log(L))$ -layer encoder that solves the sequential function composition task. For the first layer, the value of w can be shared over all positions $[n]$ using one attention head. For layer $\ell \geq 2$, one can use one attention head to retrieve the value of $2^{\ell-2}$ -hop composition value for every position. Hence, the value of i_L can be obtained using at most $2 \log_2(L) + 1$ layers of attention. The proof is similar to Theorem 4.2 of [51].¹⁴

APPENDIX B

ENCODER LOWER BOUNDS IMPLY CIRCUIT LOWER BOUNDS

In this section, we will sketch a proof that a lower bound for *encoder-only* transformers would imply breakthrough circuit lower bounds against constant-depth symmetric circuits. A constant-depth symmetric circuit is a constant depth circuit in which every gate has unbounded fan-in and computes a symmetric boolean function on its inputs. A function f is symmetric if function value $f(x)$ only depends on the number of 1's in x .

More formally, we will show that any depth- L symmetric circuit with s wires can be simulated by an encoder-only transformer of depth $6L$ and $Hdp = O((s/n)^2)$. Therefore, any lower bound of $Hdp =$

¹⁴There is one minor difference: The L functions of [51] are of the same size, whereas our L functions are of different size (after fixing the value of w). Nevertheless, one can check that the proof still can go through.

$\Omega(n^\epsilon)$ against encoders is also an $\Omega(n^{1+\epsilon/2})$ -wire lower bound against symmetric circuits.

Proving lower bounds against constant-depth symmetric circuits is notoriously hard.¹⁵ It was shown [46] that the inner product module 2 function¹⁶ requires a symmetric circuit of $\Omega(n)$ gates. However, no non-trivial lower bounds were known even for $O(n)$ -wire depth-3 symmetric circuits (see, e.g., [55]).¹⁷ Therefore, given our simulation result, if for any $\epsilon > 0$ we can prove an $\Omega(n^\epsilon)$ -size lower bound against depth-18 encoders, we would get lower bounds against $n^{1+\epsilon/2}$ -wire depth-3 symmetric circuits, which would be a breakthrough in circuit complexity theory.

On the choice of attention/MLP layers.: Here we allow the parameters for the attention and MLP layers to change arbitrarily for different positions (our autoregressive communication model applies to this case as well). Otherwise, the encoder architecture would only have $\text{poly}(Hdp)$ bits of non-uniformity; an encoder lower bound of $Hdp = \Omega(n^\epsilon)$ would then follow directly from the folklore time hierarchy theorem with advice $\text{DTIME}(n^{k+1}) \not\subseteq \text{DTIME}(n^k)/o(n)$ (see, e.g., Proposition 1 of [52]).¹⁸

Simulating a single symmetric gate.: On input $x_1, x_2, \dots, x_n \in \{0, 1\}^n$, a symmetric gate with w wires outputs $f(x_{i_1} + x_{i_2} + \dots + x_{i_w})$ for some fixed function $f: \{0, 1, 2, \dots, w\} \rightarrow \{0, 1\}$.

First, suppose each input bit x_j , we can set the V-value to be x_j and K -value to be $\log^2(n) \cdot (\vec{j}, \vec{1} - \vec{j})$ where $\vec{j} \in \{0, 1\}^d$ is the binary representation of j . Similar to the retrieval head in Section A, this allows an attention head to only attend to position j and get the value x_j .

Second, we need to gather the sum $x_{i_1} + x_{i_2} + \dots + x_{i_w}$. We set the number of attention heads $H = 5s/n$ and $d = H$. Then we divide into two cases and solve each with two layers:

- **(Handling small fan-in gates.)** If the fan-in of the symmetric gate $w \leq H$, at layer f_{attn}^1 , we gather the sum with w attention heads of a single position $k \in [n]$ of the first level. Each attention head $j \in$

¹⁵A technical reason is that a very powerful technique for proving lower bounds against constant-depth circuits, the random restriction method, fails when applying to certain symmetric functions such as the parity function.

¹⁶Given input $x_1, \dots, x_n, y_1, \dots, y_n \in \{0, 1\}^{2n}$, compute $\sum_{i=1}^n x_i \cdot y_i \pmod{2}$.

¹⁷For the case of depth-2 symmetric circuits, it was shown by [5] that E^{NP} (exponential-time with access to an NP oracle, an extremely large complexity class) requires depth-2 circuits of at least $n^{2-\epsilon}$ gates; no non-trivial results were known when we restrict the hard function to be in NP.

¹⁸We consider such a lower bound uninteresting because it does not say anything meaningful about the Transformer architecture other than it's not sufficiently non-uniform. An alternative approach of incorporating a super-linear amount of non-uniformity is to allow arbitrary positional encodings; we choose to work with position-dependent attention/MLP layers to make our presentation simpler.

$[w]$ attends to a single input position i_j using the retrieval head and get value x_{i_j} .

Afterwards, let $y_k^1 \in \mathbb{R}^{dH}$ denote the output of all H attention heads. We can choose the parameter of the MLP layer $f_{\text{mlp}}^1 : \mathbb{R}^{dH} \rightarrow \mathbb{R}^d$, so that it computes the linear sum over these attention heads and output it in an arbitrary entry t , that is $f_{\text{mlp}}(y_k^1)_t = x_{i_1} + x_{i_2} + \dots + x_{i_w}$.

In this case, we do not need a second layer. We can simply let $f_{\text{mlp}}^2 \circ f_{\text{attn}}^2$ be identity by choosing f_{attn}^2 to be retrieval heads and f_{mlp}^2 be an identity MLP.

- **(Handling large fan-in gates.)** If the fan-in of the symmetric gate $w > H$, without loss of generality we can assume that $w = H \cdot r$ for some $r \in \mathbb{Z}$ by padding and this blows up the number of wires by at most a factor of 2.

At layer $f_{\text{mlp}}^1 \circ f_{\text{attn}}^1$, we spread the sum to $w/H = r$ positions. For each $j \in [r]$, we use a position $k_j \in [n]$ to gather the partial sum $x_{i_{(j-1) \cdot H+1}} + \dots + x_{i_{j \cdot H}}$. This can be done using the same construction as the first case.

Next, we will use a single attention head in f_{attn}^2 to gather all the partial sums. Specifically, suppose i is the index of the gate we are simulating and $\vec{i}$ is its binary representation. We will set the K-value to be $(\vec{i}, \vec{1} - \vec{i})$ and V-value be the partial sums. Suppose $p = \Theta(\log n)$, following the same analysis as the retrieval head, the query $Q = (\vec{i}, \vec{1} - \vec{i})$ will only attend to these partial sums and get their average. Since r is fixed when the symmetric circuit is given, we can use f_{mlp}^2 to multiply the average by r and get $x_{i_1} + x_{i_2} + \dots + x_{i_w}$.

Thirdly, we need to compute $f(x_{i_1} + x_{i_2} + \dots + x_{i_w})$ from the sum. Here, because f is a fixed function, we can implement a look-up table for f similar to Section 3.2 of [13]:

- **(Handling small fan-in gates.)** If $w \leq H$, at layer f_{attn}^3 , we use a single attention head to retrieve the sum $x_{i_1} + x_{i_2} + \dots + x_{i_w}$. Let $y_k^3 \in \mathbb{R}^{dH}$ denote the output of all H attention heads. In MLP layer f_{mlp}^3 , we copy the sum to w different entries of $f_{\text{mlp}}^3(y_k^3) \in \mathbb{R}^d$. Then we let f_{attn}^4 and f_{attn}^5 be simple retrieval heads, so that $f_{\text{mlp}}^5 \circ f_{\text{attn}}^5 \circ f_{\text{mlp}}^4 \circ f_{\text{attn}}^4$ can simulate a two-layer MLP network. As shown in [13], such a two-layer network can compute the piecewise linear function that equals $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t]$ on the t -th entry ($t \in [w]$). Multiplying the corresponding weights by $f(t)$ gives $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t] \cdot f(t)$. Finally, we use f_{attn}^6 and f_{mlp}^6 to sum $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t] \cdot f(t)$ over $t \in [w]$ and get $f(t)$. This is similar to how we gather the partial sums.

- **(Handling large fan-in gates.)** Otherwise, when $w > H$, we again without loss of generality assume $w = H \cdot r = d \cdot r$ for $r \in \mathbb{Z}$. We again spread $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t] \cdot f(t)$ for different $t \in [w]$ to r different positions. We set up $f_{\text{mlp}}^5 \circ f_{\text{attn}}^5 \circ f_{\text{mlp}}^4 \circ f_{\text{attn}}^4 \circ f_{\text{mlp}}^3 \circ f_{\text{attn}}^3$ in the same way as in Case 1, so that the j -th ($j \in [r]$) position computes $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t] \cdot f(t)$ for all $t \in [(j-1) \cdot d + 1, j \cdot d]$.

Then we use f_{attn}^6 and f_{mlp}^6 to gather them. Let R be the set of r positions we use here. For each position $k \in R$, the output of f_{mlp}^5 is a vector $x_k^5 \in \mathbb{R}^d$ containing the value of $\mathbb{1}[x_{i_1} + x_{i_2} + \dots + x_{i_w} = t] \cdot f(t)$ for d many t 's. For f_{attn}^6 , we gather $\frac{1}{r} \sum_{k \in R} x_k$ in the same way as how we gather the partial sums using f_{attn}^2 . Then we multiply it by r and also sum over the d entries with f_{mlp}^6 . This gives the value of $f(t)$ in a single entry in the output of f_{mlp}^6 .

Simulating a symmetric circuit.: We now apply the above simulation to each gate in one layer of our symmetric circuit. For those gates with fan-in w less or equal to H , in the above construction, they only use w attention heads and at most w entries after the MLP (note $d = H = 5s/n$). As each position has H attention heads, we can pack as many as possible such gates at a single position $k \in [n]$.

After packing, no two positions will simultaneously have fan-in less than $H/2$ because otherwise we can simply pack them into a single position. In the end, all gates with fan-in less than H uses at most $s/(H/2)+1 \leq n/2$ positions.

Then for those gates with fan-in w bigger than H , our strategy of spreading it into $\lceil w/H \rceil < w/H+1$ positions will use at most $2 \cdot s/H \leq n/2$ positions as well. Thus the n positions per layer encoder we have is sufficient for the simulation. Finally, stacking the simulation of each one of the L layers together, we get a encoder with $6L$ layers and $Hdp = O((s/n)^2)$.